Literature review on Atomic and Molecular Data for NH_3 kinetics in edge plasma for EIRENE

S. Touchard, J. Mougenot, C. Rond, K. Hassouni

LSPM – CNRS UPR 3407 – Université Paris 13, Sorbonne Paris Cité, 99 Avenue J.B. Clément, 93430 Villetaneuse, France

Corresponding author : *sylvain.touchard@lspm.cnrs.fr*

Table of contents

I. Introduction

As reported in recent articles of research groups working within Eurofusion consortium [1,2,3], the use of N_2 as impurity seeds in Tokamak-devices is subject to a nitrogen balance issue: a large fraction of the nitrogen injected for the radiative cooling of the plasma is not recovered as N_2 upon regeneration using liquid helium operated cryopump. This was clearly shown and confirmed in recent global gas balance experiments at ASDEX Upgrade (AUG) and JET[1]. The most probable loss channels are the implantation of nitrogen-containing ion into plasma-facing materials, nitrogen containing species co-deposition and ammonia formation[3].

Since more than two decades, EIRENE model developed by ITER research groups, allowed reproducing the main trends of H_2/D_2 chemistry in Tokamak devices[4]. EIRENE makes use of optimized chemical kinetics model that describe both homogeneous gas phase chemistry and heterogeneous surface chemistry involved in the plasma-surface interaction (PSI). There was however no attempt to implement a chemical model in order to describe ammonia formation at the plasma-wall interface under tokamak edge-plasma conditions. In particular, the recent update of AMJUEL module used for kinetic data source for EIRENE global model does not include ammonia formation scheme [5].

The objective of this research collaboration project is to build up a reduced chemical kinetic model that can be used in EIRENE in order to reach a satisfactory description, at least in terms of orders of magnitude and trends, of the homogeneous and surface chemistry of ammonia formation in tokamak edge plasmas. Considering the large number of possible elementary processes that take place when plasmas are generated in a mixture involving nitrogen and hydrogen elements (N/H plasmas), we limited our literature survey and collisional data analysis to the following species/reactions/conditions:

- Only N, N_2, $N_2(A)$, $N_2(a')$, $N(^2D)$, NH, NH_2, NH_3, N^+, N_2^+, NH^+, NH_2^+, NH_3^+, NH_4^+ species are considered. The contribution of short living radiative states in the ground state and metastabe states kinetics was also evaluated and taken into account when necessary.
- Reactions linked to nitrogen ions N^+ to N^{7+} need not be included in the database, except for those reactions transforming N^+ into N.
- The plasma conditions considered are those corresponding to tokamak edge plasmas and relevant to ITER operation, i.e. a neutral pressure in the range 10^{-4}-20 Pa, a

plasma temperatures between 1 and 1000 eV, an electron density between 10^{18}-10^{23} m^{-3}, and a surface temperature in the range 70-2000°C. The surface of the plasma-facing material is assumed to be saturated with hydrogen and nitrogen. Of course, these conditions correspond to very wide ranges of plasma parameters. This is especially the case for electron and surface temperatures. It is therefore unlikely that a single reduced model can yield an accurate quantitative description for all the targeted range of plasma conditions.

In this report, we start by presenting the most recent studies published on the modeling of collisional processes, chemical kinetics and plasma surface interaction (PSI) in non-equilibrium N/H plasmas. Then we describe the main sources of fundamental kinetic data used in the identification of the key-processes and the development of reduced models. Taking into account the wide range of the plasma conditions of interest, we analyzed the kinetics of electrons/neutral, electron-ion, ion/neutral and neutral/neutral homogeneous processes, as well as surface processes that result in ammonia formation.

II. Research strategy

The chemistry of N/H plasmas was the subject of a large interest in four main fields[6] : Material processing, astrophysics and astrochemistry, nuclear fusion and catalytic synthesis of Ammonia from molecular nitrogen and hydrogen[7].

Material processing: N/H low pressure non-equilibrium plasmas were extensively investigated and used in order to develop metal nitriding processes for mechanical applications, i.e., hardness improvement of cutting tools [8], or semi-conductor industry, high-an low-k dielectrics[9]. The plasmas considered in this first field are laboratory plasmas with electron energy in the range 1-10 eV, electron densities in the range 10^{8}-10^{13} cm^{-3}, working pressure in the range 1-100 Pa and a gas temperature in the range 300-1000 K. Usually high energy ions are produced in these plasmas. Both homogeneous and heterogeneous (PSI) processes were considered and addressed in this first field.

Astrophysics and astrochemistry: The chemistry in gas-phase reactions alone cannot explain the abundance of gas phase H_2, NH_3 and other complex species in interstellar clouds. A combination of gas-phase and surface chemistry on ice and dust particles was therefore invoked to explain the observed abundance for a variety of chemical species[10], among which ammonia. Research in this field is nowadays very active and a theoretical effort is being devoted to improve the kinetic models[11-12], and to design and improve advanced experimental studies of surface reactions on cosmic ice and dusts[13]. It is worthy to mention here that the data derived in the context of astrophysics and astrochemistry studies corresponds to very low temperatures, i.e., 10 K<T<30 K, and species density, typically 10^{3}-10^{5} cm^{-3} in the dense region of interstellar clouds.

Nuclear fusion: the interest for N/H plasmas in this field was first motivated by the possibility to avoid a:C-H film deposition[14],[15] and subsequent tritium retention safety issue[16], when carbon was still expected as a component of ITER divertor. Preliminary laboratory plasma experiments indicated indeed that the introduction of nitrogen in C/H plasmas tends to reduce a:C-H film formation. This motivated many studies on the reduction of a-C:H film deposition by the introduction of Nitrogen using either binary or ternary mixtures of H_2, CH_4, and N_2[17,18]or NH_3[19,20]. These studies provided some insight in the relevant chemical processes in C/H/N plasmas under discharge conditions used to generate non-equilibrium laboratory plasmas.

Catalytic synthesis of ammonia from molecular nitrogen and hydrogen7: the development and optimization of the ammonia synthesis process was the main driver for a detailed investigation of surface processes involved in NH_3 catalytic formation from N_2 and H_2. The dissociation of nitrogen at the catalyst surface, i.e., dissociative chemisorption, was recognized as the limiting step for ammonia production. This motivated the use of low pressure non-equilibrium N/H plasma with the idea to provide an enhanced pre-dissociation of N_2 by either electron-impact processes or a vibrational-pumping dissociation mechanism in the gas phase. [21]

III. Summary of the most recent studies on N_2/H_2 plasma chemistry.

Although a quite significant number of articles were published on chemical models for low pressure non-equilibrium N_2/H_2 plasmas, there are only two major recent works that combine detailed experimental measurements and kinetic modeling of these plasmas.

The first one[6] is the work published by the group of the Instituto de Estructura de la Materia, IEM-CSIC in Madrid in 2011. This group derived a mechanism of NH_3 formation in a hollow DC discharge working at pressures in the range 1-10 Pa, gas and surface temperatures around 300 K, a fairly low electron density, i.e., $n_e \approx$ 1-5x10^{10} cm^{-3}, and an electron temperature in the range 1-5 eV. The mechanism they propose includes 56 homogeneous reactions, 10 heterogeneous ion-recombination processes and 15 surface chemistry processes resulting in ammonia formation. This mechanism was validated for the following plasma process conditions: low pressure (0.8–8 Pa), and a 90% H_2 + 10% N_2 initial mixture. The analysis of the measurements by mean of a kinetic model allowed identifying the main processes responsible for the observed distributions of neutrals and ions at different pressure conditions. In particular, these authors concluded that the chemistry of neutral species is dominated by the formation of a significant amount of NH_3, i.e., up to 70% of the nitrogen containing species, at the metallic wall of the reactor through the hydrogenation of atomic nitrogen and nitrogen containing radicals. They also emphasis the relative contribution of electron-impact and ion-conversion processes in the observed ion distribution.

The second study[22] published in 2015 by two German groups combines experimental measurements of neutral, radical, and ion densities, as well as chemical kinetics modeling of H_2-N_2-Ar plasmas generated in capacitively coupled radiofrequency discharges at a pressure of 1.5 Pa, input power of 200W and N_2 composition ranging between 0 and 56%. Such conditions would correspond to a fairly low plasma density, i.e., below $10^{11}cm^{-3}$, an electron temperature of few electron-volts ad gas and surface temperatures slightly above 300 K. These groups proposed a chemical model for $Ar/N_2/H_2$ plasmas with additional N/H processes as compared with former studies. In particular, reactions involving N_2-metastable states are considered. The model was validated by comparison with experiments.

If these two models are the more recently published on non-equilibrium N_2/H_2 plasmas, they both often refer to a series of paper published by Gordiets et al. in the frame of collaborations with the groups of Toulouse [23-24], Lisbon[25] and Bari[26]. In particular, Gordiets et al. [24] proposed the most quoted surface chemistry model for NH_3 formation in N_2/H_2 plasma in 1998. This model is inspired from the work of Boudart et al.[27]. The plasma chemistry model was validated for pressure-values ranging between 60 and 250 Pa, and the contribution of nitrogen-metastables, $N_2(a'^1\Sigma_u^-)$ and $N_2(A^3\Sigma_u^+)$, in the dissociation kinetics of H_2 was clearly evidenced[20].

IV. Critical description of the main kinetics data sources

The analysis of the different kinetic model of N_2/H_2 plasma published in the literature shows that one may distinguish six families of elementary processes. For each of these families, two or three main data compilations, mainly based on experimental measurements, are available.

These six main families are

- electron/neutral collisions
- electron/ion dissociative recombination
- ion/neutral collisions
- metastable-states chemistry
- neutral/neutral chemistry
- surface chemistry

We give below a short description of the different sources used to build up the data compilation we propose at the end of this report.

a. Electron-neutral processes

i. Cross section data sources for electron/neutral collision

Under low pressure and weak to moderate ionization conditions, such as those encountered in tokamak edge plasmas, the plasma kinetics is usually dominated by electron-heavy species collisional processes. Kinetic data for this type of elementary processes are quite well known for atomic species and many stable molecular species such as molecular hydrogen and nitrogen. A large effort was devoted to building up fairly detailed cross section databases for these species over a wide range of electron energy. Among the available databases, some like the electron-impact cross section databases of Janev et al [28] and Reiter[29] for molecular hydrogen, atomic hydrogen and a large number of hydrocarbon species are often quoted. They were in particular used in the model and reaction scheme derived by Amjuel[5].

Collisional processes of electron with radical species, e.g., NH, NH_2, or electronically excited species, e.g., N_2(A), N_2(a'), have been however less addressed due to the difficulty of cross section measurements for these species (beam experiments). Nevertheless, theoretical calculations were performed to estimate ionization and dissociation cross section data for electronically excited states of some diatomic molecules (H_2 [30]and N_2[31]).

Beside the initial cross-section database of Cosby et al. [32] and used in the model proposed by Amjuel, consistent electron-impact cross sections sets recommended and often updated by Phelps and co-workers[33] since the 70's are of prime importance since they yield accurate prediction of electron swarm coefficients. Also of great interest are the electron-impact cross section sets published by Hayashi [34] in 2003 and Itikawa [35] more recently.

The electron-nitrogen, electron-hydrogen and electron-atom collisions cross sections sets mentioned above were recently critically reviewed, analyzed and compiled in the frame of the LXCAT project[36]. LXCAT platform gives access to a fairly large set of cross section data for electron scattering from ground state neutral atoms and molecules (see example on NH_3

ionization process in Figure 1). These data are given over a wide energy range, i.e., from 0 to several 100's of eV and even higher for some processes. The platform gives information on total, momentum transfer, ionization, attachment, vibrational excitation (molecules) and electronic excitation cross sections. The complete set of cross sections may be used as input to a Boltzmann equation solver in order to determine the electron energy distribution function in collisional non-equilibrium plasmas. As far as electron-impact ammonia ionization is concerned, Figure 1 shows an example of output from the LXCAT platform where the cross-section data from several databases and over different energy ranges are combined for NH_3 ionization by electronic collision. This allows both comparing different sources and accessing the cross-section value over a wide range of electron energy. Note however that only one source data, i.e., Hayashi one, provides cross section data near the threshold, which is of prime importance for plasma conditions with low electron temperature values, i.e., 110 eV.

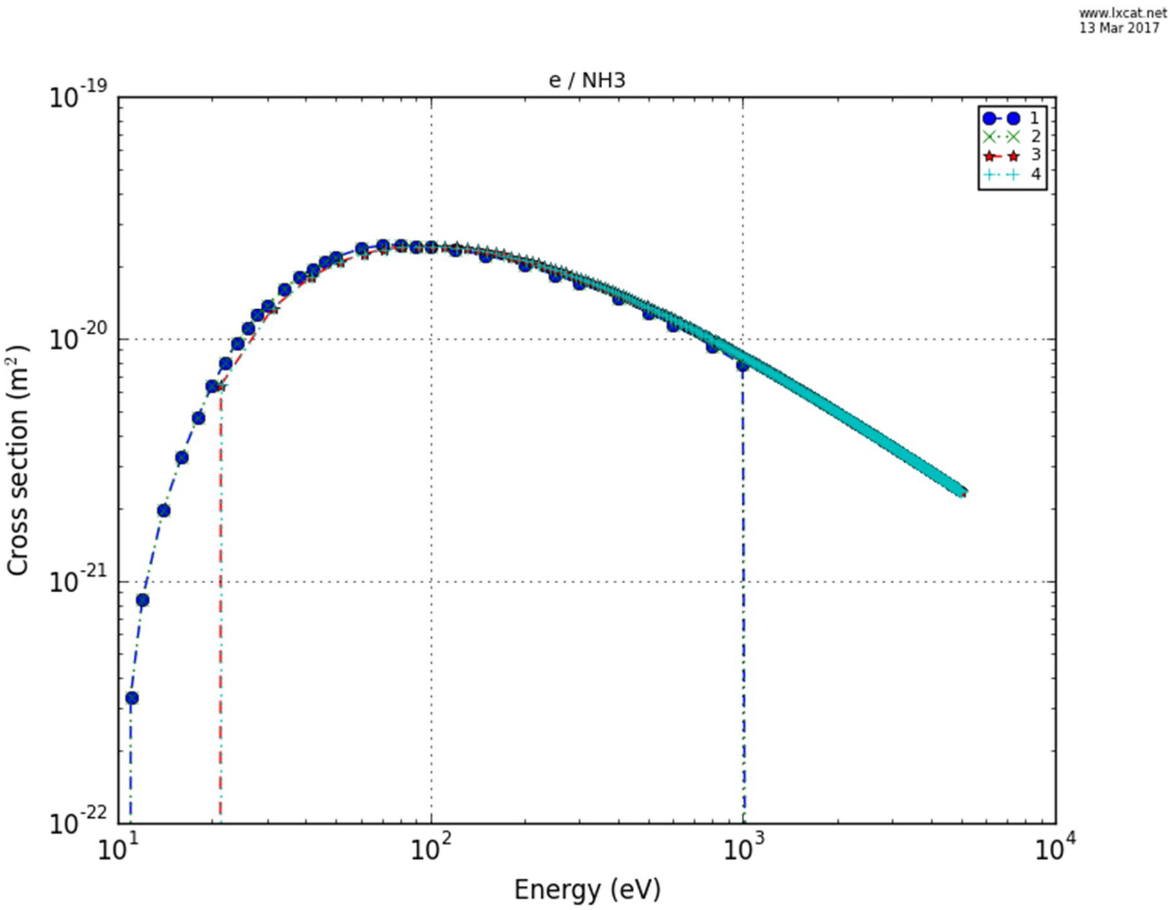

Figure 1 : Comparison between different sources cited in Lxcat Hayashi database (1), Morgan database (2), Quantemol database (3&4) for cross section in NH_3 ionization processes

ii. From electron-impact cross section data to electron-heavy species reaction rate constants

In principle, the cross-section data obtained from different database can be used to determine electron-heavy species reaction rate constants, provided the electron energy distribution function is given. Usually, the rate constants are assumed to depend only on the electron temperature and are therefore expressed as function of this parameter using dedicated curve fits.

There are many of such curve fits expressions proposed by different authors in the literature. The most common mathematical form is the generalized Arrhenius expression, as proposed by Sode et al. [22] for instance. More complex expressions involving polynomials or combination of polynomial and exponential functions are also proposed (see for example Carrasco et al.6). The existence of such a variety of expressions is simply due to the fact that the "exact" numerical estimation of a reaction rate constant using a realistic cross section data cannot be curve fitted by a simple mathematical expression over a very large range of electron temperature even if a Maxwellian EEDF is assumed. Therefore, simple curve fit expressions are generally only valid over a limited range of electron temperature. Derivation of such expressions to be used in Eirene simulations should be accompanied with a critical analysis of the expected accuracy over the considered range of electron temperature.

To illustrate the accuracy issue related to the use of simple rate constant expressions, figures 2 and 3 show 'striking' examples of the difference one may find between the estimates given by three different sources for the same rate constants for a priori two very well-known processes, i.e., H_2 and N_2 ionization processes, over a wide range of electron temperature. It appears that the rate constant expressions given by Sode et al.[22] and Amjuel [5] differ by less than one order of magnitude in the energy range 1-1000 eV for hydrogen and 5-200 eV for nitrogen. One may therefore conclude that both the Arrhenius form expressions proposed by Sode et al.[22] or Amjuel5 et al. may be used to estimate the nitrogen ionization rate constant. The accuracy of these expressions may be however quite limited for an electron temperature above 500 eV. In the case of hydrogen, the differences between the three data sources are much more significant. We would recommend the Arrhenius form proposed by Sode et al.[22] since it is in agreement with the values obtained from Amjuel5 for intermediate electron energy range on one hand, and it yields the steep decrease usually observed when numerically computing the collision integrals at the low energy range on the other hand (this is not the case for the expression proposed by Amujel5). It remains however that the accuracy of the expression proposed by Sode et al.[22] may be limited at very low and very high electron temperature.

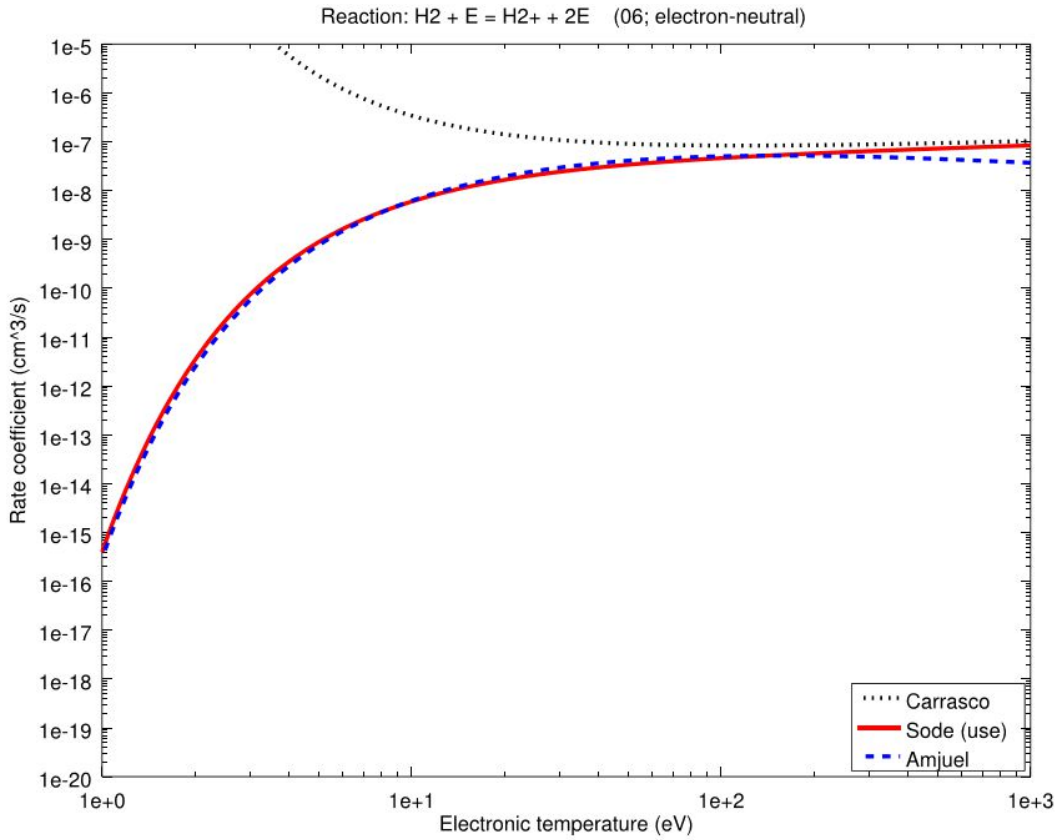

Figure 2 : Comparison of different rate coefficient data for ionization of H_2 by electron impact

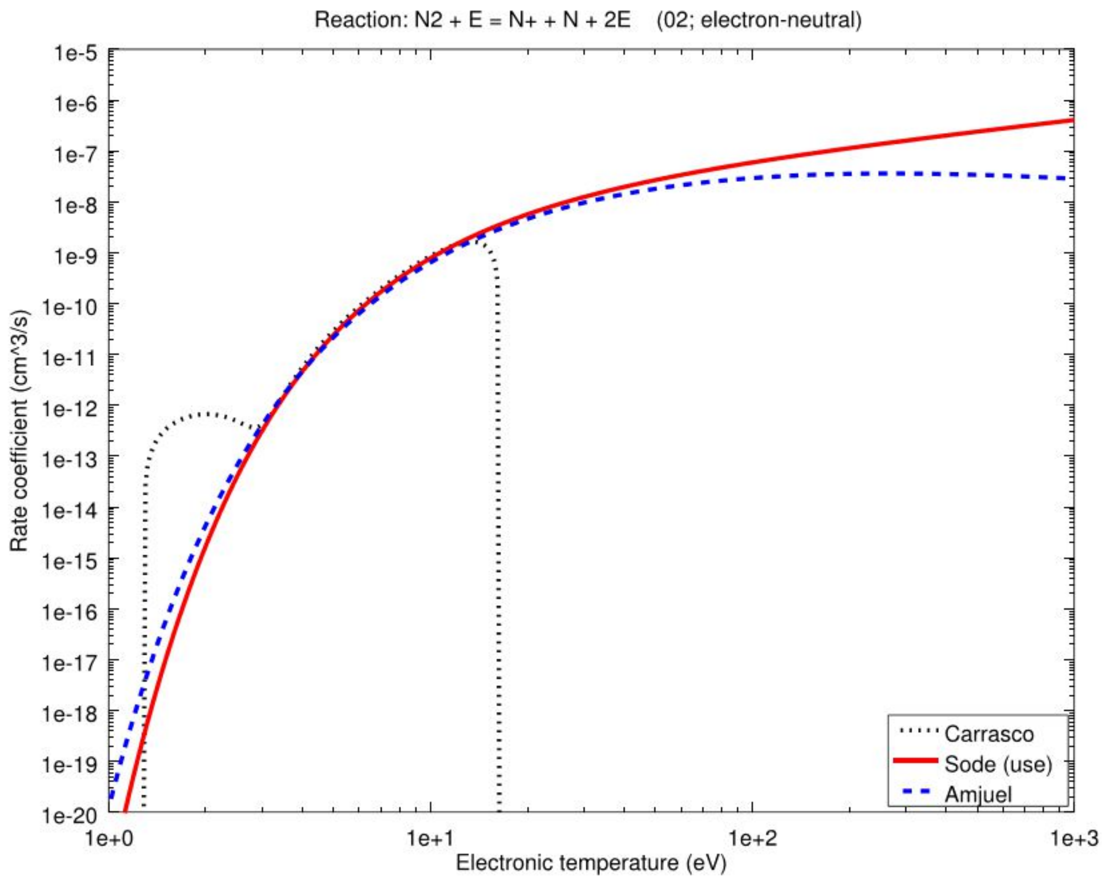

Figure 3 : Comparison of different rate coefficient data for ionization of N_2 by electron impact

iii. Rate constant data for electron-impact processes

Based on the literature survey summarized in the last sections, we identified a chemistry model that takes into account the electron-impact collisional processes reported in Table 1 and Table 2.

For the kinetic data concerning the electron impact ionization processes, we recommend the curve fit expressions proposed by Sode et al[22]. These are mainly obtained using the cross sections databases proposed by Itikawa [35].

We also suggest using generalized Arrhenius form recommended by these authors according to the expression given in equation (1), although the accuracy of these expressions may be limited at very high and very low electron energy.

$$k(T_e) = A \times T_e^n \times exp\left(-\frac{Ea}{T_e}\right) \quad (1)$$

With A in $cm^3.s^{-1}$ and Ea and Te in eV

This recommendation was not only motivated by the consistency of the data obtained from the Arrhenius expression with the existing data (and their validation by the authors under low pressure laboratory plasma conditions and for N_2 concentration close to those expected in SOL-PS), but also by the good agreement of the rate constant data estimated with these expressions and those obtained with the fitting parameter used in the current version of Eirene with H_2 (see Figure 2) and N_2 (see Figure 3).

Table 1 : Ionization processes collision

Reaction	Kinetic constant in $cm^3.s^{-1}$ / Te in eV	Reference	Appendix number
$N + e \rightarrow N^+ + 2e$	$9.3\times10^{-9} \times Te^{0.56} \times exp(-16.66/Te)$	37	1
$N_2 + e \rightarrow N^+ + N + 2e$	$2.9 \times 10^{-9} \times Te^{0.72} \times exp(-29.71/Te)$	38	2
$N_2 + e \rightarrow N_2^+ + 2e$	$1.3 \times 10^{-8} \times Te^{0.56} \times exp(-17.07/Te)$	38	3
$H + e \rightarrow H^+ + 2e$	$1.1 \times 10^{-8} \times Te^{0.29} \times exp(-15.28/Te)$	39	4
$H_2 + e \rightarrow H^+ + H + 2e$	$9.4 \times 10^{-10} \times Te^{0.45} \times exp(-29.94/Te)$	40	5
$H_2 + e \rightarrow H_2^+ + 2e$	$2.3 \times 10^{-8} \times Te^{0.19} \times exp(-17.87/Te)$	40	6
$NH + e \rightarrow NH^+ + 2e$	$2.1 \times 10^{-8} \times Te^{0.37} \times exp(-15.49/Te)$	41	7
$NH + e \rightarrow N^+ + H + 2e$	$7.6 \times 10^{-9} \times Te^{0.29} \times exp(-16.82/Te)$	41	8
$NH_2 + e \rightarrow NH_2^+ + 2e$	$1.3 \times 10^{-8} \times Te^{0.5} \times exp(-12.4/Te)$	41	9
$NH_2 + e \rightarrow NH^+ + H + 2e$	$2.2 \times 10^{-8} \times Te^{0.21} \times exp(-17.97/Te)$	41	10
$NH_3 + e \rightarrow NH_3^+ + 2e$	$1.5 \times 10^{-8} \times Te^{0.4} \times exp(-13.61/Te)$	41	11
$NH_3 + e \rightarrow NH_2^+ + H + 2e$	$1.6 \times 10^{-8} \times Te^{0.34} \times exp(-15.41/Te)$	41	12
$NH_3 + e \rightarrow NH^+ + 2H + 2e$	$5.4 \times 10^{-10} \times Te^{0.37} \times exp(-26.06/Te)$	41	ZA
$NH_3 + e \rightarrow N^+ + H + H_2 + 2e$	$8.8 \times 10^{-11} \times Te^{0.59} \times exp(-29/Te)$	42	ZB
$NH_3 + e \rightarrow H^+ + NH_2 + 2e$	$1.3 \times 10^{-10} \times Te^{0.47} \times exp(-28.55/Te)$	42	ZC

Table 2 : Electron-impact dissociation

Reaction	Kinetic constant in $cm^3.s^{-1}$ / Te in eV	Reference	Appendix number
$H_2 + e \rightarrow H + H + e$	$8.4 \times 10^{-8} \times Te^{-0.45} \times \exp(-11.18/Te)$	40	13
$N_2 + e \rightarrow N + N(^2D) + e$	$2.4 \times 10^{-8} \times Te^{0.27} \times \exp(-15.53/Te)$	38	14
$NH + e \rightarrow N + H + e$	$4.7 \times 10^{-8} \times Te^{-0.22} \times \exp(-7.69/Te)$	43	15
$NH_2 + e \rightarrow N + H_2 + e$	$1.5 \times 10^{-8} \times Te^{0.38} \times \exp(-11.44/Te)$		16
$NH_2 + e \rightarrow NH + H + e$	$4.5 \times 10^{-8} \times Te^{-0.22} \times \exp(-7.61/Te)$		17
$NH_3 + e \rightarrow NH_2 + H + e$	$4.2 \times 10^{-8} \times Te^{-0.19} \times \exp(-7.59/Te)$	44	18
$NH_3 + e \rightarrow NH + H_2 + e$	$4.1 \times 10^{-8} \times Te^{-0.26} \times \exp(-4.81/Te)$		19
$NH_3 + e \rightarrow NH + 2H + e$	$1.3 \times 10^{-8} \times Te^{0.38} \times \exp(-11.06/Te)$		ZE

b. Electron-ion recombination processes

The procedure adopted for these processes is similar to the one used in the case of electron-neutral processes. It is however worthy to mention here that unlike the activated ionization and dissociation processes, ion-recombination processes usually show monotonously decreasing cross sections as function of the electron energy. [45] They are therefore not activated, E_a≈0 eV, and their rate constants decrease with electron temperature. The n exponent in the pre-exponential factor is generally close to -0.5 for these processes [46],.

The rate constant data for these processes are readily found in up-to-date databases, such as the University of Manchester Institute of Science and Technology (UMIST) database, and the recent articles published low pressure N/H plasmas modeling and mentioned above. Most of the data published for these processes are determined experimentally at ambient temperature. Some of them were confirmed by recent theoretical calculations.[45]. The process and rate constants identified in these sources are listed in Table 3. The corresponding references are given in the “reference” column of this table.

Table 3 : Dissociative recombination processes

Reaction	Kinetic constant in $cm^3.s^{-1}$	Reference	Appendix number
$H_2^+ + e \rightarrow H + H$	$1.60 \times 10^{-8} \times (Te/0.026)^{-0.43}$	47	20
$H_3^+ + e \rightarrow 3H$	$4.36 \times 10^{-8} \times (Te/0.026)^{-0.52}$	48	21
$H_3^+ + e \rightarrow H_2 + H$	$2.34 \times 10^{-8} \times (Te/0.026)^{-0.52}$	48	22
$N_2^+ + e \rightarrow N(^2D) + N$	$1.70 \times 10^{-7} \times (Te/0.026)^{-0.3}$	49	23
$NH^+ + e \rightarrow N + H$	$4.30 \times 10^{-8} \times (Te/0.026)^{-0.5}$	47	24
$NH_2^+ + e \rightarrow NH + H$	$1.02 \times 10^{-7} \times (Te/0.026)^{-0.4}$	50	25
$NH_2^+ + e \rightarrow N + 2H$	$1.98 \times 10^{-8} \times (Te/0.026)^{-0.4}$	50	26
$HN_2^+ + e \rightarrow N_2 + H$	$1.3 \times 10^{-8} \times (Te/0.026)^{-0.84}$	51	ZF
$NH_3^+ + e \rightarrow NH + 2H$	$1.55 \times 10^{-7} \times (Te/0.026)^{-0.5}$	47	27
$NH_3^+ + e \rightarrow NH_2 + H$	$1.55 \times 10^{-7} \times (Te/0.026)^{-0.5}$	47	28
$NH_4^+ + e \rightarrow NH_3 + H$	$8.49 \times 10^{-7} \times (Te/0.026)^{-0.6}$	50	29
$NH_4^+ + e \rightarrow NH_2 + 2H$	$3.77 \times 10^{-8} \times (Te/0.026)^{-0.6}$	50	30
$N_2H^+ + e \rightarrow N_2 + H$	$5.13 \times 10^{-8} \times (Te/0.026)^{-0.72}$	52	31

c. Ion-neutral processes

The review published by Anicich et al. in 1993[53] is probably the major data source used by most of the authors during the last two decades as far as ion/neutral collisions are concerned. This review was updated by the same authors in 2003[54]. It provides a large number of data with a thorough critical analysis based on comparison between at least two or three data sources that generally rely on experimental measurements for each process.

It is however worthy to mention that most of the data recommended in Anicich review are mainly centered on works of W.T. Huntress and co-workers in the 70's[55]. To expand the number of possible data sources, we therefore compare the data recommended by Anicich with those suggested in the database of UMIST[56]. This has been developed for more than two decades. It is mainly devoted to modeling phenomena in interstellar systems. The last update of 2012 contains 6173 gas-phase reactions, involving 467 species. In this database, the temperature dependence of the rate constant is not considered and only constant values are given for most of the ion-neutral reaction rate constants. Note however that the rate constants are assumed to scale as $Tg^{-0.5}$ for some processes. This scaling is classically used to account for the interaction between the permanent dipole of the neutral species and the electrical charge of the ion. This effect is significant at low temperature, i.e., below 1000 K, and when the permanent dipole of the neutral species exceeds 0.9 Debye. It may result in a large increase of the kinetic constant at low temperature due to a 'dipole locking' phenomenon[57].

The Arrhenius form expression giving the rate constants as function of the gas temperature, Tg, are compiled in Table 4. When several data sources are available for a given reaction, their number is mentioned in the column "*Reference*". The ion/neutral processes and reaction rate data selected in this work come from Anicich et al. [53] and have been recently confirmed by studies combining experimental investigation and kinetic modeling of non-equilibrium laboratory plasmas.

Table 4 : Processes and rate coefficients for ion/neutral collisionnal processes

Reaction	Kinetic constant in $cm^3.s^{-1}$ with Tg in eV	Reference	Appendix number
$H^+ + NH_3 \rightarrow NH_3^+ + H$	$3.7 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	58	32
$H^+ + NH_2 \rightarrow NH_2^+ + H$	$2.9 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59	ZL
$H^+ + NH \rightarrow NH^+ + H$	$2.1 \times 10^{-09} \times (Tg/0.026)^{-0.5}$		ZM
$H_2^+ + H \rightarrow H_2 + H^+$	6.4×10^{-10}	60	33
$H_2^+ + H_2 \rightarrow H_3^+ + H$	2.00×10^{-09}	53 evaluated from 7 ref.	34
$H_2^+ + NH_3 \rightarrow NH_3^+ + H_2$	5.70×10^{-09}	61	35
$H_2^+ + N_2 \rightarrow N_2H^+ + H$	2.00×10^{-09}	53 evaluated from 3 ref.	36
$H_2^+ + N \rightarrow N^+ + H_2$	5.00×10^{-10}	62	ZN
$H_2^+ + N \rightarrow NH+ + H$	1.9×10^{-09}	59	ZO

$H_2^+ + NH_3 \rightarrow NH_4^+ + H$	5.00×10^{-11}	62	ZP
$H_2^+ + NH_2 \rightarrow NH_3^+ + H$	5.00×10^{-11}	62	ZQ
$H_2^+ + NH_2 \rightarrow NH_2^+ + H_2$	$2.1 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59	ZR
$H_2^+ + NH \rightarrow NH_2^+ + H$	$7.6 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59	ZS
$H_2^+ + NH \rightarrow NH^+ + H_2$	$7.6 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59	ZT
$H_3^+ + N \rightarrow NH^+ + H_2$	2.60×10^{-10}	53 evaluated from one ref of huntress	37
$H_3^+ + N \rightarrow NH_2^+ + H$	3.90×10^{-10}	53 evaluated from one ref of huntress	38
$H_3^+ + NH_3 \rightarrow NH_4^+ + H_2$	$4.40 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	53 evaluated from 5 ref	39
$H_3^+ + NH_2 \rightarrow NH_3^+ + H_2$	$1.8 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59	ZU
$H_3^+ + NH \rightarrow NH_2^+ + H_2$	$1.3 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59	ZV
$H_3^+ + N_2 \rightarrow N_2H^+ + H_2$	1.90×10^{-09}	53 evaluated from 11 ref	40
$N^+ + H_2 \rightarrow NH^+ + H$	5.00×10^{-10}	53 evaluated from 11 ref	41
$N^+ + NH_3 \rightarrow NH_2^+ + NH$	4.70×10^{-10}	53 evaluated from 3 ref	42
$N^+ + NH_3 \rightarrow NH_3^+ + N$	1.70×10^{-09}	53 evaluated from 3 ref	43
$N^+ + NH_3 \rightarrow N_2H^+ + H_2$	2.10×10^{-10}	53 evaluated from 3 ref	44
$N^+ + NH \rightarrow N_2^+ + H$	$3.7 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59	ZW
$N^+ + NH \rightarrow NH^+ + N$	$3.7 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59	ZX
$N^+ + N_2 \rightarrow N_2^+ + N$	2.00×10^{-11}	63	ZY
$N^+ + H \rightarrow H^+ + N$	2.00×10^{-10}	62	ZZ
$N^+ + NH_2 \rightarrow NH_2^+ + N$	1.0×10^{-09}	59	AB
$NH^+ + H_2 \rightarrow H_3^+ + N$	1.85×10^{-10}	53 evaluated from 3 ref	45
$NH^+ + H_2 \rightarrow NH_2^+ + H$	1.00×10^{-09}	53 evaluated from 3 ref	46

$NH^+ + NH_3 \rightarrow NH_3^+ + NH$	1.80×10^{-09}	53 evaluated from 3 ref	47
$NH^+ + NH_3 \rightarrow NH_4^+ + N$	6.00×10^{-10}	53 evaluated from 3 ref	48
$NH^+ + NH_2 \rightarrow NH_3+ + N$	$1.5 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59	AC
$NH^+ + NH_2 \rightarrow NH_2^+ + NH$	1.80×10^{-09}	62	AD
$NH^+ + NH \rightarrow NH_2^+ + N$	1.0×10^{-09}	59	AE
$NH^+ + N_2 \rightarrow N_2H^+ + N$	6.50×10^{-10}	64	49
$NH^+ + N \rightarrow N_2^+ + H$	1.3×10^{-09}	59	AG
$NH_2^+ + H_2 \rightarrow NH_3^+ + H$	2.00×10^{-10}	53 evaluated from 3 ref	50
$NH_2^+ + NH_3 \rightarrow NH_3^+ + NH_2$	1.20×10^{-09}	53 evaluated from 3 ref	51
$NH_2^+ + NH_3 \rightarrow NH_4^+ + NH$	1.20×10^{-09}	53 evaluated from 3 ref	52
$NH + NH_2^+ \rightarrow NH_3^+ + N$	$7.3 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59	AH
$NH_2^+ + N \rightarrow HN_2^+ + H$	9.1×10^{-11}	59	AI
$NH_3^+ + NH_3 \rightarrow NH_4^+ + NH_2$	2.10×10^{-09}	53 evaluated from 3 ref	53
$NH_3^+ + H_2 \rightarrow NH_4^+ + H$	4.40×10^{-13}	48 evaluated from 11 ref	AJ
$NH_3^+ + NH \rightarrow NH_4^+ + N$	$7.1 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59	AK
$N_2^+ + H_2 \rightarrow NH_2^+ + H$	2.00×10^{-09}	53 evaluated from 3 ref	54
$N_2^+ + NH_3 \rightarrow N_2 + NH_3^+$	2.00×10^{-09}	53 evaluated from 3 ref	55
$N_2^+ + N \rightarrow N^+ + N_2$	1.00×10^{-11}	53 evaluated from 3 ref	AL
$N_2H^+ + NH_3 \rightarrow NH_4^+ + N_2$	2.30×10^{-09}	53 evaluated from 3 ref	56
$N_2H^+ + H_2 \rightarrow H_3^+ + N_2$	5.10×10^{-18}	53 evaluated from 3 ref	AM

d. Metastable-states chemistry

The long-term work carried out by Gordiets in collaboration with the groups of Toulouse, Lisbon and Bari, on the kinetic modeling of N_2/H_2 non-equilibrium microwave plasmas is certainly the most relevant studies as far as plasma-specific processes, involving vibrationally and electronically excited species, are concerned. The chemistry models developed by these authors were at least partly validated under laboratory plasma conditions, i.e., $n_e=10^8\text{-}10^{12}$ cm^{-3}, T_e=1-5 eV, T_g=300-1000 K, p=10-200 Pa and MWP_{inp}=100-1000 W. Of special interest is the paper published by Gordiets and Lisbon group in 2005 [65] and where the key-role that nitrogen metastables play in the chemistry of N_2/H_2 mixtures under the investigated plasma conditions is highlighted. A reduced set of reactions involving these metastables is proposed.

The metastable states of N and N_2, are produced by electron-impact processes. The corresponding rate constant data were derived by Sode et al.[22] using the cross sections data of Itikawa[35]. They were curve-fitted as function of the electron temperature using Arrhenius form expressions.

It is worthy to mention that this review was limited to four main metastable states: N(^{2}D), N(^{2}P), $N_2(A^3\Sigma_u^+)$ (mentioned as N_2(A)) and $N_2(a'^1\Sigma_u^-)$ (named as N_2(a')). This choice is justified by the very low pressure conditions in edge plasmas, which results in a rapid radiative de-excitation process of the other excited species that are lost before undergoing any collisional process.

Table 5 : Electronic collisional processes and rate coefficients for the formation of N and N_2 metastable states

Reactions	Kinetic constant in cm^3.s^{-1} / Te in eV	Ref.	Appendix number
$N_2 + e \rightarrow N_2(A) + e$	$1.4 \times 10^{-08} \times \exp(-6.325\times 10^{-04}/Te)$	35	EK
$N_2 + e \rightarrow N_2(a') + e$	$5.1 \times 10^{-09} \times \exp(-0.001/Te)$	35	EL
$N + e \rightarrow N(^2D) + e$	$1.0 \times 10^{-8} \times Te^{-0.36} \times \exp(-7.15\times 10^{-05}/Te)$	35	EM
$N + e \rightarrow N(^2P) + e$	$5.5 \times 10^{-15} \times Te^{-0.41} \times \exp(-9.05\times 10^{-05}/Te)$	35	EN

Beside the work of Gordiets on N_2/H_2 plasmas, reactions of atomic and molecular nitrogen metastables, i.e., N(^{2}D), N(^{2}P) and $N_2(A^3\Sigma_u^+)$, with stable molecules, such as H_2, N_2, NO, N_2O, CO, CO_2, CH_4 or C_2H_4, were reviewed in 1999 by Herron et al.[66]. This review is of great interest since the author's recommendation is based on a critical comparison between at least 2 or 3 experimental measurements for each process.

In both Gordiets et al.[23] article, and Herron et al.[66] review, the temperature dependence of the rate constants is generally not considered for metastable quenching processes, and only single constant values are given.

On the basis of these two data compilation sets, the processes we identified for the metastable states of N_2 and N are listed in Table 6. We choose to use the same kinetic data as Gordiets, since these values are within the margin of error of Herron's recommendations and rely on well identified experimental measurements, the references of which are given in the "reference" column of Table 6.

Table 6 : Processes and rate coefficients for quenching of N_2 metastable states by neutrals

Reactions	Kinetic constant in $cm^3.s^{-1}$ / Tg in eV	Ref.
$N_2(A)+N(^4S) \rightarrow N_2(X)+N(^4S)$	2.00×10^{-12}	67
$N_2(A)+N(^4S) \rightarrow N_2(X)+N(^2P)$	$1.80 \times 10^{-09} \times (Tg/0.026)^{-0.67}$	68 69 70
$N_2(A)+N_2(X) \rightarrow 2N_2(X)$	3.01×10^{-10}	71
$N_2(A)+NH_3 \rightarrow N_2(X)+H+NH_2$	8.51×10^{-11}	72
$N_2(A)+H \rightarrow N_2(X)+H$	2.09×10^{-10}	73
$N_2(A)+H_2 \rightarrow N_2(X) + 2H$	$2.00 \times 10^{-10} \times \exp(-0.3016/Tg)$	7475
$N_2(A)+N_2(A) \rightarrow N_2(X)+N_2(B)$	3.01×10^{-10}	76-77
$N_2(A)+N_2(A) \rightarrow N_2(X)+N_2(C)$	1.50×10^{-10}	78
$N_2(A)+N_2(X, v > 6) \rightarrow N_2(B)+N_2(X, v- 6)$	3.01×10^{-11}	79
$N_2(a')+N_2(X) \rightarrow N_2(B)+N_2(X)$	1.90×10^{-13}	80-81
$N_2(a') + H \rightarrow N_2(X)+H$	1.50×10^{-10}	23
$N_2(a') + H_2 \rightarrow N_2(X) + 2H$	2.61×10^{-11}	81
$N_2(a') + M \rightarrow N_2(a)+M$	$1.80 \times 10^{-11} \times \exp(-0.1464/Tg)$	82
$N_2 (a)+M \rightarrow N_2(a') +M$	9.00×10^{-12}	82
$N_2(v > 16) + N_2(v >= 16) \rightarrow N_2(a')+N_2(X)$[a]	$2.00 \times 10^{-14} \times \exp(-0.1465/Tg)$	23

Table 7 : Processes and rate coefficients for quenching of N metastable states by neutrals.

Reactions	Kinetic constant in $cm^3.s^{-1}$ / Tg in eV	Ref.
$N(^2D)+N_2(X) \rightarrow N_2(X)+N(^4S)$[b]	6.00×10^{-09}	46, 83
$N(^2D)+H_2 \rightarrow NH + H$	2.29×10^{-12}	84
$N(^2D)+NH_3 \rightarrow NH+NH_2$	1.10×10^{-10}	83
$N(^2P)+N(^4S) \rightarrow N(^2D)+N(^4S)$	1.79×10^{-12}	46
$N(^2P)+N_2 (X) \rightarrow N(^4S)+N_2(X)$	1.99×10^{-18}	85
$N(^2P)+H_2 \rightarrow NH + H$	2.51×10^{-14}	86
$N+N+N_2 \rightarrow N_2(A)+N_2$	1.02×10^{-09}	87,88 89
$N+N+N \rightarrow N_2(A)+N$	1.02×10^{-14}	

[a] $N_2(X)$ is the ground state of N_2
[b] $N(^4S)$ is the ground state of N

e. Neutral/neutral chemistry

For the reactions between ground state neutral species produced in N/H plasmas, a global comprehensive mechanism such as those proposed by Capitelli et al.[25] in their book published in 2000 can include up to 82 reactions. However, since our objective is to build up a simplified mechanism that mainly describes low pressure plasma conditions, we suggest to use the mechanism proposed by Gordiets et al. [23]This mechanism contains 17 processes and was validated and adopted by many research groups working on non-equilibrium low pressure plasmas.

Note however that we have checked and confirmed this mechanism and the related collisional data by comparison with data recommended in two other databases: the NIST database available online[90] and the database of Baulch et al.[91], the last update of which date back to 2005

Finally, to be consistent with our choice for metastable states reactions, we selected the processes given in Gordiets short kinetic scheme except for the three body reactions that were not considered due to the very low pressure conditions. The neutral/neutral processes and corresponding rate constant data that were selected are listed in Table 8.

Table 8 : Processes and rate coefficients for neutral/neutral reactions

Reaction	Kinetic constant in $cm^3.s^{-1}$ / Tg in eV	Reference	Appendix number
$H_2 + N \rightarrow NH + H$	$4.65 \times 10^{-11} \times \exp(-1.4304/Tg)$	92	AQ
$N + NH \rightarrow N_2 + H$	5.0×10^{-11}	93	AW
$H + NH \rightarrow N + H_2$	$5.4 \times 10^{-11} \times \exp(-0.0142/Tg)$	94	AV
$NH + NH \rightarrow N_2 + H_2$	$5\times 10^{-14} \times (Tg/0.026)^{+1.0}$	94	
$NH + NH \rightarrow NH_2 + N$	$1.7 \times 10^{-12} \times (Tg/0.026)^{+1.5}$	97	AY
$NH + NH \rightarrow N_2 + 2H$	8.5×10^{-11}	95	AX
$H + NH_2 \rightarrow NH + H_2$	$6.6 \times 10^{-11} \times \exp(-0.1586/Tg)$	94	AT
$N + NH_2 \rightarrow N_2 + H$	1.2×10^{-10}	93	
$N + NH_2 \rightarrow N_2 + H_2$	1.2×10^{-10}	93	
$NH + NH_2 \rightarrow NH_3 + N$	1.66×10^{-12}	97	
$H_2 + NH_2 \rightarrow NH_3 + H$	$5.4 \times 10^{-11} \times \exp(-0.5594/Tg)$	96	AR
$H + NH_3 \rightarrow NH_2 + H_2$	$8.4 \times 10^{-14} \times (Tg/0.026)^{+4.1} \times \exp(-0.4102/Tg)$	97	AU
$H_2 + NH \rightarrow NH_2 + H$	$5.96 \times 10^{-11} \times \exp(-0.6706/Tg)$	98	AS

f. Surface reactions

There are basically three groups of reactions as far as surface chemistry under N_2/H_2 plasma conditions is concerned. These are: (i) Wall recombination of positive ions, (ii) metastable de-excitation at the wall and (iii) neutral chemistry and ammonia formation

(i) Wall recombination of positive ions

As suggested by many authors, we choose for this type of processes to follow the classical theory of wall-recombination of charged species as described in the reference book of M. A. Lieberman and A. J. Lichtenberg [99] . Ions reaching the wall are supposed to recombine with electron of the wall with a probability of 1. Usually the recombination is dissociative and the resulting neutral fragments are assumed to be immediately remitted in the gas with a probability of 1. The kinetic scheme for this type of process is listed in Table 9

Table 9 : Kinetic scheme for ion neutralization on metallic wall

Wall neutralisation processes	
$H^+ + wall \rightarrow H$	$NH^+ + wall \rightarrow N + H$
$H_2^+ + wall \rightarrow H + H$	$NH_2^+ + wall \rightarrow NH + H$
$H_3^+ + wall \rightarrow H_2 + H$	$NH_3^+ + wall \rightarrow NH_2 + H$
$N^+ + wall \rightarrow N$	$NH_4^+ + wall \rightarrow NH_3 + H$
$N_2^+ + wall \rightarrow N + N$	$N_2H^+ + wall \rightarrow N_2 + H^+$

(ii) Metastable Wall-relaxation

The wall-de-excitation of metastable states may be more or less effective depending on the ability of the molecule-metal system to relax the fairly high amount of energy produced during the de-excitation process. Surprisingly, a short literature survey shows that the wall de-excitation probability of metastables is pressure dependent and may vary from 10^{-3} at atmospheric pressure to 1 at low pressure. Since we are mainly interested in very low pressure conditions, we recommend a de-excitation coefficient of 1 for both $N_2(A)$ metastables. We would also recommend to assume dissociative de-excitation for $N_2(A)$.

It is more difficult to state on the value of the de-excitation probability of $N(^2D)$ at the wall. As a matter of fact, the metastable-metal system does not show a simple metastable energy relaxation route during metastable-wall interaction in this case. The de-excitation probability of $N(^2D)$ may be therefore smaller than 1. Nevertheless, in the absence of any even qualitative information on the wall-de-excitation of this metastable, we would suggest to assume a probability of 1 for its de-excitation.

(iii) Surface chemistry of neutral species and NH_3 formation mechanism

The chemistry of ammonia production by heterogeneous catalytic reaction between N_2 and H_2 was thoroughly investigated during the 70's by surface science, catalysis and industrial chemistry communities (ammonia synthesis was a key-chemical industrial process at that time) [100]. Among the many groups that addressed this process, the long term thorough and very high quality studies of Ertl [100]et al. and Somorjai et al. [101]enabled shedding the light on the dynamic and kinetics of the gas-surface interaction processes involved in the formation of ammonia on a variety of model mono-crystalline iron surfaces. It is now widely accepted that ammonia formation proceeds through two classical mechanisms: the Langmuir Hinshelwood mechanism, where ammonia formation results from the reaction between two adsorbed species and the Eley-Rideal mechanism where ammonia formation proceeds through a successive reaction between several gas phase H-atom and nitrogen containing adsorbed species.

Beside the fundamental studies mentioned above, a thorough investigation of the catalytic efficiency of different materials was carried out by the group of Venugopalan from Western Illinois University[102]. This group clearly showed that the catalytic efficiency of ammonia formation varies with the nature of the wall. The group reported the following trends of catalytic efficiency for different materials: Pt> Ag > Fe> Cu> Al> Zn.

Nevertheless, to our best knowledge, there has been no attempt in the literature to transpose the pioneering work of Ertl et al.[100] and Somorjai et al.[101] on the detailed mechanism of ammonia formation to non-equilibrium plasma conditions. Most of the studies on ammonia formation in plasmas were conducted under post discharge conditions, in the absence of any ion bombardment that may significantly affect surface processes, which makes the conclusion drawn on the basis of post-discharge experiments not completely transferable to plasma conditions.

In any case among the mechanisms proposed in the literature, we focused on three main groups of sources:

- The Langmuir Hinshelwood mechanism proposed by Ertl and co-worker[100]
- The detailed surface chemistry of Gordiets and co-workers[24]
- The simplified surface mechanism suggested by Carrasco and al. [6]

The first group of articles corresponds to the previously mentioned long-term work of Gerhard Ertl (The 2007 Nobel Prize in Chemistry)[103] on N_2, H_2 and NH_3 surface chemistry[100]. This work was summarized in a book published in 1991, where the surface chemistry of H_2 and N_2 on iron surface was thoroughly analyzed. The authors also showed how the ammonia formation mechanism and related kinetic data change with the pressure, the wall temperature and the crystallographic orientation of the surface. The major conclusions of this work are:

- N_2 may weakly adsorb in two adsorbed states with adsorption energies of 21 kJ.mol^{-1} and 31 kJ.mol^{-1} and sticking coefficients of 0.5 and 0.01, respectively.
- Dissociative chemisorption probability of molecular nitrogen on zero coverage monocrystalline surface range between 10^{-7} and $5x10^{-6}$ depending on the crystallographic orientation of the surface.
- Due to this low chemisorption probability, the Nitrogen adsorption is the limiting step as far ammonia catalytic production is concerned.
- Dissociative Chemisorption of N_2 leads to a strongly bonded N-atom, the thermal desorption of which only occurs above 700 K. Also, the mobility of N-atom becomes appreciable only above 500 K.

- At zero coverage, the sticking coefficient for hydrogen-molecule dissociative adsorption is equal to 0.1, which is more than five orders of magnitude larger than in the case of nitrogen.
- The variation of hydrogen sticking coefficient as a function of the surface coverage Θ, follows a Langmuir-type two-site adsorption process: $S=S_0 \times (1-\Theta)^2$
- Surface mobility and desorption kinetics of H-atom are also much larger than their counterparts for nitrogen. Basically, desorption is completed at 500 K. Above this temperature, the surface coverage by H ad-atom is governed by adsorption-desorption equilibrium.
- Ammonia may form following a L-H mechanism by three successive hydrogenation of an almost immobile N ad-atom by 3 highly mobile H ad-atom even at fairly low temperature, i.e., below 500 K, provided nitrogen ad-atom can be formed.
- Adsorption energy of NH_3 on Fe is around 70 $kJ.mol^{-1}$
- Ammonia may decompose to form adsorbed NH at the surface even at fairly moderate temperature: 350-400 K.

Note that the conclusions listed above would indicate that under plasma environment, one may expect E-R production of ammonia through hydrogenation of adsorbed N, NH and NH_2 by gas phase H-atom.

On the basis of the above findings, Ertl and coworkers derived an ammonia production model based on the Langmuir Hinshelwood mechanism shown on the energy diagram depicted in Figure 4. This mechanism involves a two-step dissociative adsorption process for N_2, a one-step dissociative adsorption process for H_2, 3 successive hydrogenation processes of adsorbed N-atom and desorption of ammonia.

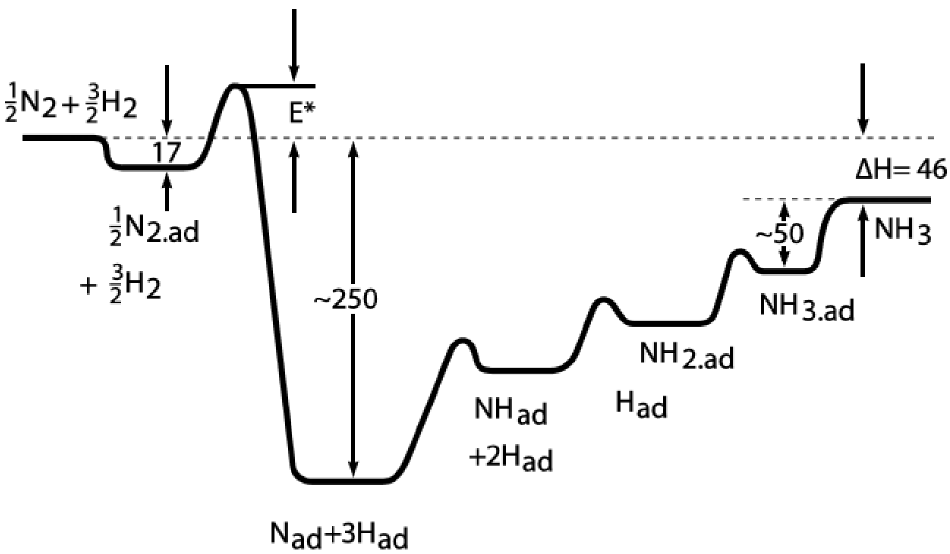

Figure 4 Energy diagram showing the progression of the reaction from the reactants N and H to the product NH. Energies are given in units of kJ/mol. (from Ertl 1991[100])

The kinetic values suggested by Ertl for the different steps of the Langmuir Hinshelwood mechanism are available in Table 10. The kinetics of adsorption processes for gas phase species is described in terms of an initial zero-coverage sticking coefficient, γ, the variation of which as a function of the wall-temperature is also given by an Arrhenius form expression:

$$\boldsymbol{\gamma = \gamma 0 \times \exp\left(-\frac{E_a}{T_w}\right)}$$

Where $\gamma 0$ and E_a (in K) are the pre-exponential factor and the activation energy of the adsorption process, respectively. Tw is the wall temperature.

The kinetics of reactions between adsorbed species is described in terms of reaction frequencies in s^{-1}.

One of the difficulties that arise when using Ertl model comes from the lack of data for hydrogenation processes that involve two ad-atoms and lead to the formation of NH_3. This difficulty may be circumvented by considering that these processes are very fast and that ammonia formation is essentially limited by adsorption processes. This seems to be the option implicitly proposed by Ertl et al.

An alternative option, is to use the approach of Kim and Boudart [27], who suggest that the reaction frequency between two species adsorbed on a metallic surface depends on :

- the diffusion frequency of an atom on the metallic surface υ_D. This is typically around $10^{13} s^{-1}$, a value that corresponds to the surface phonon frequency according to the theory of surface diffusion proposed by Barth[104]
- the activation temperature, E_d/R, for the surface diffusion of H-atom is approximately 2300 K.
- the activation temperature, E_r/R, of the reaction between the two absorbed species

The reaction frequency between two adsorbed species is given by:

$$\nu = \frac{\nu_D}{4} \times \exp\left(-\frac{E_D + \boldsymbol{E_r}}{T_w}\right)$$

The activation temperatures for the diffusion process of H-atom and N-atom on iron are approximately 2300 K and 10400 K, respectively. The activation energy of NH_{ads} and $NH_{2\text{-}ads}$ by hydrogen ad-atoms are obtained from the energy diagram reported by Ertl and depicted in figure 4. The NH_{ads} and $NH_{2\text{-}ads}$ hydrogenation processes shows activation energy of 0.3 eV and 0.2 eV, respectively.

If the new species formed by diffusion on the surface is desorbed, a third factor is involved in the computation of the desorption kinetic rate: the desorption frequency of an atom from the metallic surface υ_d. This factor is typically around $10^{15} s^{-1}$ according to Kim and Boudart [27]

Table 10: Sticking coefficient and surface adsorption energy for surface processes occurring in the Langmuir-Hinsheldwood mechanism proposed by Ertl[100] as a function of wall temperature (T_W)

Surface Processes	Surface type	Initial sticking coefficient γ or reaction frequency ν (s^{-1}) with Tw in eV	Adsorption energy E_{ads} (in eV)	Adsorption energy E_{ads} (in K)
$N_2 \rightarrow N_{2ad}$	Fe	1.0×10^{-2}	0.176	2045
$N_{2ad} \rightarrow N_2$	Fe	$1.0 \times 10^{10}.\exp(-0.3213/Tw)$	0.176	2045
$N_{2ad} \rightarrow 2N_{ad}$	Fe	$1.0\times10^{7} \exp(-0.2902/Tw)$	2.695	31273
$H_{2ad} \rightarrow 2H_{ad}$		$0.1 \exp(-0.0311/Tw)$	0.913-1.130	10600-13110
$H_{ad}+N_{ad} \rightarrow NH_{ad}$	Fe	Not given Expression may be derived from Boudart with $\nu_D=10^{13}$ s^{-1}, E_d=0.2 eV and E_r=0.3 eV	1.1	12750
$H_{ad}+NH_{ad} \rightarrow NH_{2ad}$	Fe	Not given Expression may be derived from Boudart with $\nu_D=10^{13}$ s^{-1}, E_d=0.2 eV and E_r=0.3 eV	0.342	3970
$H_{ad}+NH_{2ad} \rightarrow NH_{3ad}$	Fe	Expression may be derived from Boudart with $\nu_D=10^{13}$ s^{-1}, E_d=0.2 eV and E_r=0.3 eV	0.425	4931
$NH_3 \rightarrow NH_{3\ ad}$	Fe	0.16	0.518	6014

One of the major drawbacks of the model proposed by Ertl and coworkers is that it does not account for a possible Eley-Rideal (E-R) process. This process may be very significant as far as catalytic NH_3 formation is concerned if H-atom shows very high density in the gas phase, which is usually the case under plasma conditions. A detailed NH_3 formation mechanism involving 23 surface reactions that include E-R processes was proposed by Gordiets et al.[24]. In principle, this model, the reactions of which are listed in Table 11, enables one describing the ammonia production over a wide range of surface temperature and for a variety of gas phase composition. Unfortunately, this model involves a large number of parameters that are hardly accessible. The value reported in Table 11, were fitted using loss coefficient measurements performed in low pressure non-equilibrium N_2H_2 flowing microwave plasmas.

Table 11 : Global model proposed by Gordiets et al.[24] and sticking coefficient for surface processes

	Reaction	Initial sticking coefficient on free surface	Energy(K)
S1	$N + F_V \rightarrow Nf$	Not given	Not given
S2	$H + Fv \rightarrow Hf$	Not given	Not given
S3	$N + Sv \rightarrow N(s)$	Not given	0
S4	$Nf + Sv \rightarrow N(s)+Fv$	Not given	Not given
S5	$N + N(s) \rightarrow N_2 + Sv$	6.0×10^{-5}	900
S6	$Nf + N(s) \rightarrow N_2 + Fv + Sv$	Not given	2600
S7	$H + Sv \rightarrow H(s)$	Not given	0
S8	$Hf + Sv \rightarrow H(s) + Fv$	Not given	500
S9	$H + H(s) \rightarrow H_2 + Sv$	1.2×10^{-2}	840
S10	$H + H(s) \rightarrow H_2 + Fv + Sv$	Not given	2340
S11	$N + H(s) \rightarrow NH(s)$	Not given	Not given
S12	$Nf + H(s) \rightarrow NH(s) + Fv$	Not given	4000
S13	$H + N(s) \rightarrow NH(s)$	Not given	Not given
S14	$Hf + N(s) \rightarrow NH(s) + Fv$	Not given	4000
S15	$NH + Sv \rightarrow NH(s)$	Not given	3000
S16	$NH_2 + Sv \rightarrow NH_2(s)$	Not given	3000
S17	$H + NH(s) \rightarrow NH_2 (s)$	Not given	Not given
S18	$Hf + NH(s) \rightarrow NH_2 (s) + Fv$	Not given	0
S19	$NH + H(s) \rightarrow NH_2 (s)$	1.0×10^{-2}	3000
S20	$NH_2 (s) + H \rightarrow NH_3 + Sv$	Not given	Not given
S21	$NH_2 (s) + Hf \rightarrow NH3 + Sv + Fv$	Not given	0
S22	$H+NH_2(s) \rightarrow NH_3 + Sv$	Not given	3000
S23	$H_2 + NH(s) \rightarrow NH_3 + Fs$	Not given	0

Table 12 : Simplified scheme for surface processes proposed by Carrasco et al. [6,105]

Reaction	Sticking Coefficient or reaction frequency (s^{-1})	Origin of the value
H+Fs -> H(s)	1	99
H + H(s) -> H_2 +Fs	1.5×10^{-3}	106
N + Fs -> N(s)	1	99
N + N(s) -> N_2 +Fs	6.0×10^{-3}	99
NH + Fs -> NH (s)	1	99
NH_2 + Fs -> NH_2 (s)	1	99
N + H(s) -> NH(s)	1×10^{-2}	24
H + N(s) -> NH(s)	8.0×10^{-3}	24
H + NH(s) -> NH_2 (s)	8.0×10^{-3}	24
NH + H(s) -> NH_2 (s)	1×10^{-2}	24
NH(s) + H(s) -> NH_2 (s) + Fs	10^{13}*exp(-0.5/T_w) T_w in eV	107, 108
H+NH_2(s) -> NH_3 +Fs	8.0×10^{-3}	109
NH_2 + H(s) -> NH_3 +Fs	1×10^{-2}	24
H_2 + NH(s) -> NH_3 +Fs	8.0×10^{-4}	24
NH_2 (s) + H(s) -> NH_3 +2Fs	10^{13}*exp(-0.4/Tw) T_w in eV	107,108

A simplified version of this model was proposed by Carrasco et al.[6]. This model listed in Table 12 also account for both LH and ER mechanisms, and essentially relies on Ertl's and Gordiet's mechanisms. The data used for surface reaction rates are based on the work of Ertl and using the surface diffusion theory described above in this section. This model was validated by comparison with experimental results in N_2/H_2 plasmas at low pressure (0.2 to 8 bar) and with low N_2 concentration. This is why we recommend it for EIRENE simulations.

V- A reduced kinetic scheme for NH_3 formation mechanism under conditions of edge plasmas

As we discussed in the previous sections of this report, taking into account all the elementary processes reported for N_2/H_2 plasmas can lead to a kinetic scheme involving more than two hundred reactions. However, since the required computation cost for models such as EIRENE may vary exponentially with the number of species and reactions, such a large kinetic model is not practical and should be therefore reduced.

Therefore, we first performed an approximate estimation of the characteristic times of the different processes. Taking into account the wide targeted ranges of plasma parameters and wall temperature, we choose to setup a kinetic scheme that ensures that each species generated in the plasma can be consumed in the gas phase. We therefore identify in a first step (step 1) the major gas phase production and consumption process for each species that were included in the model. This implies that the number of processes should be at least two times larger than the number of species, i.e., which means that we should have at least 30 processes. Of course, wall-recombination of ions, wall-de-excitation of metastables and surface chemistry for neutrals is also taken into account.

Then, in a second step (step 2) we estimated the diffusion characteristic times to the surface for the conditions of interest. We disregarded all the processes with characteristic times larger than the diffusion one at the upper limit value of the pressure. Almost all the processes involving only neutral ground state species can be disregarded on the basis of this consideration.

Then the characteristic times of the production and consumption process for each species are estimated assuming nitrogen abundance of less than 4% (2% N_2 in the plasma). We also assume that nitrogen containing species show similar densities. Hydrogen-ions, i.e., H^+, H_2^+ and H_3^+, were also assumed to have similar densities. Then, in step 3, only processes with characteristic times comparable to the predominant production and consumption processes identified in step 1 are selected.

Table 13 : Example of comparative calculation of the reaction times of the NH_3^+ production processes

Processes type	Rate constant at Te or Tgaz	Processes frequency (in s^{-1})	Time between each processes (in s)
H^++NH_3 (Tgaz=300K)	3.7×10^{-09}	$3.7 \times 10^{+03}$	2.70×10^{-04}
H_2^+ + NH_3 (Tgaz=300K)	2.0×10^{-09}	$2.0 \times 10^{+03}$	5.00×10^{-04}
e-+NH_3 (Te=1 eV)	1.8×10^{-14}	1.8×10^{-02}	$5.43 \times 10^{+01}$
e-+NH_3(10 eV)	9.7×10^{-09}	$9.7 \times 10^{+03}$	1.04×10^{-04}
e-+NH_3(100 eV)	8.3×10^{-08}	$8.3 \times 10^{+04}$	1.21×10^{-05}
e-+NH_3(1000 eV)	2.4×10^{-07}	$2.4 \times 10^{+05}$	4.26×10^{-06}

We found that, very often, due to the wide range of electron temperature, at least 2-3 production processes have to be considered for each species.

As far as ionization processes are concerned, we considered classical electron impact dissociative and non-dissociative ionization of molecular hydrogen and nitrogen as well as ionization of atomic hydrogen and nitrogen. We found that NH_3^+, NH_2^+ and NH^+ ions are mainly produced by electron-impact at high electron temperature, i.e., T_e>10 eV and by charge transfer from hydrogen ions at low electron temperature, i.e., T_e<10 eV. It is therefore important to consider both electron-impact and charge transfer processes in order to achieve accurate description of ammonia ionization over the very wide range of electron temperature considered in this work. We showed however that electron-impact ionization on NH_2 radical can be disregarded, NH_2^+ being mainly obtained by dissociative ionization of NH_3.

The dissociation of ammonia into neutral fragments through electron-impact processes was found to be a possible major potential source of NH_2 and NH radical even at low electron temperature. The corresponding processes were therefore included in the proposed model. We showed however that dissociation of NH_2 and NH can be disregarded.

There is a large number of ion conversion processes. These may be of great importance in determining nitrogen-containing ion, N^+, NH^+, NH_2^+, NH_3^+, NH_4^+ and N_2H^+ populations distribution, which may in turn affect the neutral nitrogen containing species. Taking into account the large ranges of electron temperature and gas pressure, we consider that the populations of the three hydrogen ions, H^+, H_2^+ and H_3^+, may be significant, depending on the actual plasma conditions. We therefore take into account all the conversion processes involving hydrogen ions. These processes are very fast, i.e., their characteristic times are very small. They are likely to govern hydrogen ions population at low electron temperature and high pressure, and nitrogen ion populations at low electron temperature. Due to the low abundance of nitrogen, N/H < 4×10^{-2}, the ion conversion process involving two nitrogen-containing ions are disregarded.

As for nitrogen metastable, we considered only $N_2(A)$ and $N(^2D)$ states. We found that they are mainly produced by electron-impact processes. They are lost by collisional quenching with H-atom, H_2-molecule and NH_3. Of course metastable wall-de-excitation is also taken into account.

As a result of the analysis summarized in the last paragraphs, we found that N/H plasmas chemistry may be described using 50 gas phase reactions. A further reduction to 40 reactions scheme can be performed with additional assumptions, including removal of the metastable chemistry and ions lumping. Even further reduction may be performed if information on the predominant hydrogen ion among H_3^+, H_2^+ and H^+ is available. As a matter of fact, ion conversion processes show very similar rate constants, which prevent any possible reduction among these processes without any information on the predominance of the species involved in this process.

To conclude, we recommend a reduced mechanism involving the 50 gas phase processes, reported Table 14 to Table 20, and the 15 surface reactions proposed by Carrasco and listed in Table 12. It is possible to further reduce this mechanism to 36 gas phase reactions and 15 surface reactions mechanism by lumping the reactions in light-gray rows Table 16 and Table 17 and disregarding the dark gray reactions involving the metastable-states of the atomic and molecular nitrogen (Table 18 to Table 20).

Table 14 : Reduced kinetic scheme for ionization processes by collision with electron

Number in Appendix	Reaction	Kinetic constant in $cm^3.s^{-1}$ / Te in eV	Reference
1	$N + e \rightarrow N^+ + 2e$	$9.3\times10^{-9} \times Te^{0.56} \times \exp(-16.66/Te)$	37
2	$N_2 + e \rightarrow N^+ + N + 2e$	$2.9 \times 10^{-9} \times Te^{0.72} \times \exp(-29.71/Te)$	38
3	$N_2 + e \rightarrow N_2^+ + 2e$	$1.3 \times 10^{-8} \times Te^{0.56} \times \exp(-17.07/Te)$	38
4	$H + e \rightarrow H^+ + 2e$	$1.1 \times 10^{-8} \times Te^{0.29} \times \exp(-15.28/Te)$	39
5	$H_2 + e \rightarrow H^+ + H + 2e$	$9.4 \times 10^{-10} \times Te^{0.45} \times \exp(-29.94/Te)$	40
6	$H_2 + e \rightarrow H_2^+ + 2e$	$2.3 \times 10^{-8} \times Te^{0.19} \times \exp(-17.87/Te)$	40
7	$NH + e \rightarrow NH^+ + 2e$	$2.1 \times 10^{-8} \times Te^{0.37} \times \exp(-15.49/Te)$	41
11	$NH_3 + e \rightarrow NH_3^+ + 2e$	$1.5 \times 10^{-8} \times Te^{0.4} \times \exp(-13.61/Te)$	41
12	$NH_3 + e \rightarrow NH_2^+ + H + 2e$	$1.6 \times 10^{-8} \times Te^{0.34} \times \exp(-15.41/Te)$	41

Table 15 : Reduced kinetic scheme for Electron-impact dissociation

Number in Appendix	Reaction	Kinetic constant in $cm^3.s^{-1}$ / Te in eV	Reference
13	$H_2 + e \rightarrow H + H + e$	$8.4 \times 10^{-8} \times Te^{-0.45} \times \exp(-11.18/Te)$	40
14	$N_2 + e \rightarrow N + N(^2D) + e$	$2.4 \times 10^{-8} \times Te^{0.27} \times \exp(-15.53/Te)$	38
18	$NH_3 + e \rightarrow NH_2 + H + e$	$4.2 \times 10^{-8} \times Te^{-0.19} \times \exp(-7.59/Te)$	44
ZE	$NH_3 + e \rightarrow NH + 2H + e$	$1.3 \times 10^{-8} \times Te^{0.38} \times \exp(-11.06/Te)$	

Table 16 : Reduced kinetic scheme for Dissociative recombination processes*

Number in Appendix	Reaction	Kinetic constant in $cm^3.s^{-1}$ / Te in eV	Reference
20	$H_2^+ + e \rightarrow H + H$	$1.60 \times 10^{-8} \times (Te/0.026)^{-0.43}$	47
21	$H_3^+ + e \rightarrow 3H$	$4.36 \times 10^{-8} \times (Te/0.026)^{-0.52}$	*48*
22	$H_3^+ + e \rightarrow H_2 + H$	$2.34 \times 10^{-8} \times (Te/0.026)^{-0.52}$	*45*
23	$N_2^+ + e \rightarrow N(^2D) + N$	$1.70 \times 10^{-7} \times (Te/0.026)^{-0.3}$	49
24	$NH^+ + e \rightarrow N + H$	$4.30 \times 10^{-8} \times (Te/0.026)^{-0.5}$	47
25	$NH_2^+ + e \rightarrow NH + H$	$1.02 \times 10^{-7} \times (Te/0.026)^{-0.4}$	50
27	$NH_3^+ + e \rightarrow NH + 2H$	$1.55 \times 10^{-7} \times (Te/0.026)^{-0.5}$	*47*
28	$NH_3^+ + e \rightarrow NH_2 + H$	$1.55 \times 10^{-7} \times (Te/0.026)^{-0.5}$	*47*
29	$NH_4^+ + e \rightarrow NH_3 + H$	$8.49 \times 10^{-7} \times (Te/0.026)^{-0.6}$	50
31	$N_2H^+ + e \rightarrow N_2 + H$	$5.13 \times 10^{-8} \times (Te/0.026)^{-0.72}$	52

* Processes that may be lumped : 21 with 22, and 27 with 28.

Table 17 : Reduced kinetic scheme for ion/neutral collisionnal processes

Number in Appendix	Reaction	Kinetic constant in $cm^3.s^{-1}$ / Tg in eV	Reference
32	$H^+ + NH_3 \rightarrow NH_3^+ + H$	$3.7 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	58
ZL	$H^+ + NH_2 \rightarrow NH_2^+ + H$	$2.9 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
ZM	$H^+ + NH \rightarrow NH^+ + H$	$2.1 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	
33	$H_2^+ + H \rightarrow H_2 + H^+$	6.4×10^{-10}	60
34	$H_2^+ + H_2 \rightarrow H_3^+ + H$	2.00×10^{-09}	53
35	$H_2^+ + NH_3 \rightarrow NH_3^+ + H_2$	5.70×10^{-09}	61
36	$H_2^+ + N_2 \rightarrow N_2H^+ + H$	2.00×10^{-09}	53
ZN	$H_2^+ + N \rightarrow N^+ + H_2$	5.00×10^{-10}	62
ZO	$H_2^+ + N \rightarrow NH+ + H$	1.9×10^{-09}	59
ZR	$H_2^+ + NH_2 \rightarrow NH_2^+ + H_2$	$2.1 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
ZT	$H_2^+ + NH \rightarrow NH^+ + H_2$	$7.6 \times 10^{-10} \times (Tg/0.026)^{-0.5}$	59
39	$H_3^+ + NH_3 \rightarrow NH_4^+ + H_2$	$4.40 \times 10^{-09} \times (Tg/300)^{-0.5}$	53
ZU	$H_3^+ + NH_2 \rightarrow NH_3^+ + H_2$	$1.8 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
ZV	$H_3^+ + NH \rightarrow NH_2^+ + H_2$	$1.3 \times 10^{-09} \times (Tg/0.026)^{-0.5}$	59
40	$H_3^+ + N_2 \rightarrow N_2H^+ + H_2$	1.90×10^{-09}	53
41	$N^+ + H_2 \rightarrow NH^+ + H$	5.00×10^{-10}	53
ZZ	$N^+ + H \rightarrow H^+ + N$	2.00×10^{-10}	62
46	$NH^+ + H_2 \rightarrow NH_2^+ + H$	1.00×10^{-09}	53
50	$NH_2^+ + H_2 \rightarrow NH_3^+ + H$	2.00×10^{-10}	53

* Processes that may be lumped: 32 with 35 ; ZL with ZR and ZU ; ZM with ZT and ZV.

Table 18 : Electronic collisional processes and rate coefficients for the formation of N and N_2 metastable states

Number in Appendix	Reactions	Kinetic constant in $cm^3.s^{-1}$ / Te in eV	Ref.
EK	$N_2 + e \rightarrow N_2(A) + e$	$1.4 \times 10^{-08} \times \exp(-6.325\times 10^{-04}/Te)$	35
EM	$N + e \rightarrow N(^2D) + e$	$1.0 \times 10^{-8} \times Te^{-0.36} \times \exp(-7.15\times 10^{-05}/Te)$	35

Table 19 : Processes and rate coefficients for the quenching and the excitation of N_2 electronic levels by heavy particle collisions

Reaction of main species with N_2 metastables			
Number in Appendix	Reactions	Kinetic constant in $cm^3.s^{-1}$ / Tg in eV	Ref.
DB	$N_2(A)+N(^4S) \rightarrow N_2(X)+N(^2P)$	$1.80 \times 10^{-09} \times (Tg/0.026)^{-0.67}$	68-70
DD	$N_2(A)+NH_3 \rightarrow N_2(X)+H+NH_2$	8.51×10^{-11}	72
DE	$N_2(A)+H \rightarrow N_2(X)+H$	2.09×10^{-10}	73
DF	$N_2(A)+H_2 \rightarrow N_2(X) + 2H$	$2.00 \times 10^{-10} \times \exp(-0.3016/Tg)$	74,75

Table 20 :Rate coefficients for the quenching of metastable nitrogen atoms by collisions with atoms and molecules.

Reaction of main species with N metastables			
Number in Appendix	Reactions	Kinetic constant in $cm^3.s^{-1}$ / Tg in eV	Ref.
EA	$N(^2D)+H_2 \rightarrow NH + H$	2.29×10^{-12}	84
EC	$N(^2D)+NH_3 \rightarrow NH+NH_2$	1.10×10^{-10}	83

<u>Important note</u>: The figures showing the evolution of the collision frequency as a function of Te, Tg or Tw of all the processes described in this report were plotted in the appendix. All reactions are identified by a reference number reported in the column named "Number in appendix" for each table. For most of the reactions, 2 or 3 data sources are compared

.

Bibliography

[1] *Plasma–wall interactions with nitrogen seeding in all-metal fusion devices: Formation of nitrides and ammonia* Martin Oberkofler , D. Alegre , F. Aumayr , S. Brezinsek , T. Dittmar , K. Dobes D. Douai , A. Drenik , M. Köppen , U. Kruezi , Ch. Linsmeier , C.P. Lungu , G. Meisl , M. Mozetic , C. Porosnicu , V. Rohde , S.G. Romanelli , the ASDEX Upgrade Team and JET Contributor Fusion Engineering and Design **98–99** (2015) 1371–1374

[2] *Nitrogen retention mechanisms in tokamaks with beryllium and tungsten plasma-facing surfaces* M Oberkofler, G Meisl, A Hakola, A Drenik, D Alegre, S Brezinsek, R Craven, T Dittmar, T Keenan, S. G. Romanelli, R Smith, D Douai, A Herrmann, K Krieger, U Kruezi, G Liang, Ch Linsmeier, M Mozetic, V Rohde, the ASDEX Upgrade team, the EUROfusion MST1 Team and 9 JET Contributors Phys. Scr. **T167** (2016) 014077 (p1-7)

[3] *Ammonia formation and W coatings interaction with deuterium/ nitrogen plasmas in the linear device GyM* L. Laguardia , R. Caniello , A. Cremona , D. Dellasega , F. Dell'Era , F. Ghezzi , G. Gittini , G. Granucci , V. Mellera , D. Minelli , F. Pallotta , M. Passoni , D. Ricci , E. Vassallo Journal of Nuclear Materials **463** (2015) 680–683

[4] *Progress in two-dimensional plasma edge modelling* D. Reiter Journal of Nuclear Materials **196-198** (1992) p.80-89

[5] *The data file AMJUEL: Additional Atomic and Molecular Data for EIRENE* D. Reiter Version: March 8, 2016 available on http://www.eirene.de/amjuel.pdf.

[6] *Neutral and ion chemistry in low pressure dc plasmas of H_2 /N_2 mixtures: routes for the efficient production of NH_3 and NH_4^+* Esther Carrasco, Miguel Jimenez-Redondo, Isabel Tanarro and Vıctor J. Herrero Phys. Chem Chem Phys., **13** (2011) 19561-19572

[7] *Ammonia synthesis catalyst 100 years: Practice, enlightenment and challenge* Huazhang Liu Chinese Journal of Catalysis **35** (2014) p.1619–1640

[8] *Diagnostic and modeling of N_2-H_2 discharges for iron nitriding*, A. Ricard, B. F. Gordiets, M. J. Pinheiro, C. M. Ferreira, G. Baravian, J. Amorim, S. Bockel, and H. Michel Eur. Phys. J. AP **4** (1998) p.87-93

[9] *Surface reactions during etching of organic low- k films by plasmas of N_2 and H_2* Kenji Ishikawa, Yoshikazu Yamaoka, Moritaka Nakamura, Yuichi Yamazaki, Satoshi Yamasaki, Yasushi Ishikawa and Seiji Samukawa Journal of Applied Physics **99** (2006) 083305

[10] *The chemistry of interstellar space* E. Herbst Chem. Soc. Rev. **30** (2001) p. 168-176.

[11] *Accurate rate coefficients for models of interstellar gas-grain chemistry* I. Lohmar, J. Krug and O. Biham, Astron. Astrophys., **504** (2009) L5-L8.

[12] *Complex molecule formation in grain mantles* P. Hall and T. J. Millar Astron. Astrophys **517**(2010) A1.

[13] *Ice surface reactions: A key to chemical evolution in space* Naoki Watanabe, Akira Kouchi Progress in Surface Science, **83** (2008) p.439-489

[14] *Plasma-material interactions in current tokamaks and their implications for next step fusion reactors* G. Federici, C. H. Skinner, J. N. Brooks, J. P. Coad, C. Grisolia, A. A. Haasz, A. Hassanein, V. Philipps, C. S. Pitcher, J. Roth, W. R. Wampler and D. G. Whyte, Nucl. Fusion, **41**, (2001) p.1967-2137

[15] W. Jacob, C. Hopf, M. Meier and T. Schwarz-Selinger, in Nuclear Fusion Research: Understanding Plasma-Surface Interactions, ed. R. E. H. Clark and D. Reiter, Springer Series in Chemical Physics **78** Springer Verlag, (2005)

[16] *Recent analysis of key plasma wall interactions issues for ITER* J. Roth, E. Tsitrone, A. Loarte, Th. Loarer, G. Counsell, R. Neu, V. Philipps, S. Brezinsek, M. Lehnen, P. Coad, Ch. Grisolia, K. Schmid, K. Krieger, A. Kallenbach, B. Lipschultz, R. Doerner, R. Causey, V. Alimov, W. Shu, O. Ogorodnikova, A. Kirschner, G. Federici and A. Kukushkin, J. Nucl. Mater. **390–391** (2009) p.1-9

[17] *Growth and erosion of amorphous carbon (a-C:H) films by low-temperature laboratory plasmas containing H and N mixtures* T. Schwarz-Selinger, C. Hopf, C. Sun and W. Jacob, J. Nucl. Mater. **363** (2007) p174-178

[18] *Ion Chemistry in Cold Plasmas of H_2 with CH_4 and N_2* I. Tanarro, V. J. Herrero, A. M. Islyaikin, I. Mendez, F. L. Tabares and D. Tafalla, J. Phys. Chem. A **111** (2007) p 9003–9012.

[19] *Suppression of Tritium Retention in Remote Areas of ITER by Nonperturbative Reactive Gas Injection* F. L. Tabares, J. A. Ferreira, A. Ramos, G. van Rooij, J. Westerhout, R. Al, J. Rapp, A. Drenik and M. Mozetic, Phys. Rev. Lett. **105** (2010) n°175006.

[20] *Suppression of hydrogenated carbon film deposition by scavenger techniques and their application to the tritium inventory control of fusion devices* F. L. Tabares, D. Tafalla, I. Tanarro, V. J. Herrero, A. Islyaikin and C. Maffiotte, Plasma Phys. Controlled Fusion 44 (2002) L37-L42.

[21] *Plasma Catalytic Synthesis of Ammonia Using Functionalized-Carbon Coatings in an Atmospheric-Pressure Non-equilibrium Discharge* J. Hong, M. Aramesh, O. Shimoni, D. H. Seo, S. Yick, A. Greig C. Charles, S. Prawer, A. B. Murphy Plasma Chemistry and Plasma Processing (2016) **36 (**4) pp 917–940

[22] *Measurement and modeling of neutral, radical, and ion densities in H_2 -N_2 -Ar plasmas* M. Sode, W. Jacob, T. Schwarz-Selinger, and H. Kersten Journal of applied physics **117** (2015) 083303

[23] *Self-consistent kinetic model of low-pressure N_2–H_2 flowing discharges: I. Volume processes* B. Gordiets, C.M. Ferreira, M.J. Pinheiro and A. Ricard, Plasma Sources Sci. Technol. **7** (1998) p.363–378

[24] *Self-consistent kinetic model of low-pressure N_2–H_2 flowing discharges: II. Surface processes and densities of N, H, NH_3 species* B Gordiets, C M Ferreira, M J Pinheiro and A Ricard, Plasma Sources Sci. Technol. **7** (1998) p.379–388

[25] *Surface kinetics of N and O atoms in N_2 –O_2 discharges* B Gordiets, C M Ferreira, J Nahorny, D Pagnon, M Touzeau and M Vialle

[26] *Plasma Kinetics in Atmospheric Gases* M. Capitelli, C.M. Ferreira, B.F. Gordiets, A.I. Osipov (2000), Springer-Verlag Berlin Heidelberg

[27] *Recombination of oxygen, nitrogen, and hydrogen atoms on silica: kinetics and mechanism* Y. C. Kim and M. Boudart Langmuir **7** (12) (1991) pp 2999–3005

[28] Elementary Processes in Hydrogen-Helium Plasmas Cross Sections and Reaction Rate Coefficients R.K. Janev, W.D. Langer, K.J. Evans., D.E.Post Jr. Springer Series on Atoms+Plasmas, **Vol 4**, 1987

[29] D. Reiter, in "Atomic and Plasma Material Interaction Processes", eds. R.K. Janev, H.W. Drawin, Elsevier Science Publishers (1993)

[30] *Towards a Cross Section Database of Excited Atomic and Molecular Hydrogen* R Celiberto, M Capitelli and A Laricchiuta Physica Scripta (2002) T96

[31] *Ionization of excited nitrogen molecules by electron impact* A.V. Kosarim, B.M. Smirnov, M. Capitelli, R. Celiberto, G. Petrella, A. Laricchiuta Chemical Physics Letters **414** (2005) p.215-221

[32] *Electron-impact dissociation of nitrogen* P.C. Cosby J. Chem. Phys **98** (1993) p.9544-9553

[33] *Cross sections and swarm coefficients for nitrogen ions and neutrals in N 2 and argon ions and neutrals in Ar for energies from 0.1 eV to 10 keV* A. V. Phelps J. Phys. Chem. Ref. Data **20** (1991) p.557

[34] M.Hayashi, *Bibliography of Electron and Photon Cross Sections with Atoms and Molecules* Published in the 20th Century-Nitrogen MoleculeNIFS-DATA-77 National Institute for Fusion Science, 2003.

[35] *Cross Sections for Electron Collisions with Nitrogen Molecules* Y. Itikawa J. Phys. Chem. Ref. Data **35** (2006) p31-53

[36] *The LXCat project: Electron scattering cross sections and swarm parameters for low temperature plasma modeling* S. Pancheshnyi, S. Biagi c, M.C. Bordage, G.J.M. Hagelaar, W.L. Morgan, A.V. Phelps, L.C. Pitchford Chemical Physics **398** (2012) 148–153

[37] *Ionization of carbon, nitrogen, and oxygen by electron impact* Y.-K. Kim and J.-P. Desclaux, Phys. Rev. A **66**, (2002) 012708.

[38] *Cross sections for electron collisions with nitrogen molecules* Y. Itikawa Journal of Physical and Chemical Reference Data **35** (2006) p.31–53

[39] *Pulsed crossed-beam study of the ionisation of atomic hydrogen by electron impact* M. B. Shah, D. S. Elliott, and H. B. Gilbody Journal of Physics B: Atomic and Molecular Physics **20** (1987) p.3501–3514.

[40] *Cross sections for electron collisions with hydrogen molecules* J.-S. Yoon, M.-Y. Song, J.-M. Han, S. H. Hwang, W.-S. Chang, B. Lee, and Y. Itikawa J. Phys. Chem. Ref. Data **37** (2008) p.913–931.

[41] *Cross-sections for the electron impact ionization of* ND_x *(x =1-3)* V. Tarnovsky, H. Deutsch, and K. Becker International Journal of Mass Spectrometry and Ion Processes **167/168** (1997) p. 69 –78

[42] *Ionization of ammonia and deuterated ammonia by electron impact from threshold up to 180 eV* T. D. Märk, F. Egger and M. Cheret The Journal of Chemical Physics **67** (1977) p.3795–3802

[43] *Etching of low-k materials for microelectronics applications by means of a* N_2/H_2 *plasma: modeling and experimental investigation* K. V. Laer, S. Tinck, V. Samara, J. F. de Marneffe, and A. Bogaerts Plasma Sources Sci. Technol. **22** (2013) 025011.

[44] *Boltzmann equation analysis of electron-molecule collision cross sections in water vapor and ammonia* M. Yousfi and M. D. Benabdessadok Journal of Applied Physics **80** (1996) p.6619–6630

[45] *Dissociative recombination of* N_2^+*: An ab initio study* D. A. Little, K. Chakrabarti, J. Zs. Mezei, I. F. Schneider, and J. Tennyson Phys. Rev. A **90** (2014) 052705

[46] *Kinetic scheme of the non-equilibrium discharge in nitrogen-oxygen mixtures* I A Kossyi, A Yu Kostinsky, A A Matveyev and V P Silakov Plasma Sources Sci.Technol. **1** (1992) p.207-220

[47] *The dissociative recombination of molecular ions* J. Brian and A. Mitchell **186** (1990) p 215-248

[48] *Dissociative recombination of rotationally cold* H_3^+ B. J. McCall, A. J. Huneycutt, R. J. Saykally, N. Djuric, G. H. Dunn, J. Semaniak, O. Novotny, A. Al-Khalili, A. Ehlerding, F. Hellberg, S. Kalhori, A. Neau, R. D. Thomas, A. Paal, F. Österdahl, and M. Larsson Phys. Rev. A **70** (2004) 052716

[49] *Dissociative recombination and excitation of* N_2^+*: Cross sections and product branching ratios* J. R. Peterson, A. Le Padellec, H. Danared, G. H. Dunn, M. Larsson, A. Larson, R. Peverall, C. Strömholm and S. Rosén M. af Ugglas W. J. van der Zande Journal of Chemical Physics **108** (1998) p.1978-1988

[50] *Branching fractions in the dissociative recombination of* $NH_4(+)$ *and* $NH_2(+)$ *molecular ions* L. Vikor, A. Al-Khalili, H. Danared, N. Djuric, G. H. Dunn, M. Larsson, A. Le Padellec, S. Rosaen, M. Af Ugglas Astronomy and Astrophysics **344** (1999) p.1027-1033

[51] *Reassessment of the dissociative recombination of N_2H^+ at crying* E. Vigren, V. Zhaunerchyk, M. Hamberg, M. Kaminska, J. Semaniak, M. af Ugglas, M. Larsson, R. D. Thomas, and W. D. Geppert The Astrophysical Journal **757** (2012) A34

[52] *The dissociative recombination rate coefficients of H_3^+, HN_2^+, and HCO^+* T. Amano Journal of Chemical Physics **92** (1990) p.6492-6501

[53] Evaluated *Bimolecular Ion-Molecule Gas Phase Kinetics of Positive Ions for Use in Modeling Planetary Atmospheres, Cometary Comae, and Interstellar Clouds* Vincent G. Anicich J. Phys. Chem. Ref. Data **22** (1993) 1469-1569.

[54] *An Index of the Literature for Bimolecular Gas Phase Cation-Molecule Reaction Kinetics* Anicich V G JPL Publication 2003, 03-19 NASA (2003)

[55] *Ion-molecule reactions in the evolution of simple organic molecules in interstellar clouds and planetary atmospheres* W.T. Huntress Chemical Society Reviews. **6** (1977) p.295-323.

[56] *The UMIST database for astrochemistry 2012* D. McElroy, C. Walsh, A.J. Markwick, M.A.Cordiner, K. Smith and T.J. Millar Astronomy & Astrophysics **550** (2013) paper A36

[57] *Statistical adiabatic channel model for ion–molecule capture processes* J. Troe J. Chem. Phys. **87** **(**1987) p2773-2780

[58] *A selected ion-flow tube study of the reactions of O^+, H^+ and HeH^+ with several molecular gases at 300 K, David Smith, Patrik Spanel and Christhopher A. Mayhew* International Journal of Mass Spectrometry and Ion Processes **117** (1992) p.457-473

[59] *A MODEL FOR GAS PHASE CHEMISTRY IN INTERSTELLAR CLOUDS: I. THE BASIC MODEL, LIBRARY OF CHEMICAL REACTIONS, AND CHEMISTRY AMONG C, N, AND O COMPOUNDS* S. S. Prasad and W. T. Huntress Jr Astrophysical Journal Supplement Series **43** (1980) p.1-35

[60] *An ion cyclotron resonance study of reactions of ions with hydrogen atoms* Karpas, Z., Anicich V.G., Huntress, W.T., J. Chem. Phys., **70**, (1979) 2877

[61] *Ion cyclotron resonance studies on the reaction of H ions with various simple molecules and hydrocarbons* J. K. Kim and W. T. Huntress, Jr., J. Chem. Phys. **62** (1975) 2820-2825.

[62] *H_2 generation in Ar/NH_3 microdischarges* R. A. Arakoni, A. N. Bhoj, and M. J. Kushner Journal of Physics D: Applied Physics **40** (2007) p2476.

[63] *Ionized physical vapor deposition of titanium nitride: A global plasma model* K. Tao, D. Mao, and J. Hopwood, Journal of Applied Physics **91** (2002) p.4040–4048

[64] *An experimental survey of the reactions of NHn+ ions (n = 0 to 4) with several diatomic and polyatomic molecules at 300 K* Adams, N.G., Smith, D., and Paulson, J.F. J. Chem. Phys. **72** (1980) p.288

[65] *Molecular dissociation in N_2–H_2 microwave discharges* E Tatarova , F. M. Dias , B Gordiets and C M Ferreira Plasma Sources Sci. Technol. **14** (2005) p.19–31

[66] *Evaluated Chemical Kinetics Data for Reactions of N(^{2}D), N(^{2}P) and $N_2\left(A^3\Sigma_u^+\right)$ in the Gas Phase* John T. Herron J. Phys. Chem. Ref. Data **28** (1999) p. 1453-1483

[67] A review mechanisms of atomic in non-equilibrium and molecular gases excitation up to 20000 K F.R. Gilmore; E. Bauer and J.W. McGowan J. Quant. Spectrosc. Radiat. Transfer. 9 (1969) 157-183

[68] *Excitation Mechanisms of the Nitrogen First-Positive and First-Negative Radiation at High Temperature* R.C. Flagan and J.P. Appleton J. Chem. Phys. **56** (1972) p1163-1173

[69] Study of the products of the reactions of $N_2(A^3\Sigma_u^+)$: The effect of vibrational energy in N_2(A) M.F. Golde and A.M. Moyle Chemical Physics Letters **117** (1985) p.375-380

[70] *The excitation of N(^2P) by N_2(A $^3\Sigma^+_u$, v'=0,1)* Piper L.G. J. Chem. Phys. **90** (1989) p.7087-7095

[71] *Energy transfer reactions of N_2(A $^3\Sigma^+_u$). 5. Quenching by hydrogen halides, methyl halides, and other molecules* W. G. Clark, D. W. Setser J. Phys. Chem **84 (18)** (1980) p.2225–2233

[72] *Energy-transfer reaction of N_2(A $^3\Sigma^+_u$) to sulfur monoxide and other diatomic and polyatomic molecules* D.Z. Cao and D.W. Setser J. Phys.Chem **92 (5)** (1988) pp 1169–1178

[73] *Experimental study of the reactions of N_2($A^3\Sigma^+_u$) with H atoms and OH radicals* G.H. Ho and M. F. Golde The Journal of Chemical Physics **95** (1991) p.8866-8870

[74] *Temperature-dependent N_2($A^3\Sigma^+_u$) quenching rate coefficients* T.G. Slanger, B.J. Wood and G. Black J. Photochem. **2** (1973/74) p.63-66

[75] *Ab initio study of electronic energy transfer in the quenching of N_2($A^3\Sigma^+_u$) and N_2($B^3\Pi_g$) by H_2* R. F. Sperlein and M. F. Golde J. Chem. Phys. **89** (1988) p.3113-3123

[76] *Reaction rate constant for $2N_2(A^3\Sigma^+_u) \rightarrow N_2(C^3\Pi_u)+N_2(X^1\Sigma^+_g, v'> 0)$* G. N. Hays, and H. J. Oskam The Journal of Chemical Physics **59** (1973) p.6088-6091

[77] *State-to-state N_2($A^3\Sigma^+_u$) energy pooling reactions. II. The formation and quenching of N_2($B^3\Pi_g$, v'=1–12)* Lawrence G. Piper The Journal of Chemical Physics **88** (1988) p6911-6921

[78] *State-to-state N_2($A^3\Sigma^+_u$) energy-pooling reactions. I. The formation of N_2($C^3\Pi_u$) and the Herman infrared system* Lawrence G. Piper The Journal of Chemical Physics **88** (1988) p231-239

[79] *The excitation of N_2($B^3\Pi_g$,v=1–12) in the reaction between N_2($A^3\Sigma^+_u$) and N_2(X,v≥5)* L. G. Piper The Journal of Chemical Physics **91** (1989) p.864-873

[80] *The deactivation of $N_2B^3\Pi_g$, v=0–2 and $N_2a'^1\Sigma^-_u$, v = 0 by nitrogen* J.W. Dreyer and D. Perner Chem. Phys. Lett. **16** (1972) p.169-173

[81] *Quenching rate coefficients for N_2($a'^1\Sigma^-_u$)* L. G. Piper the Journal of Chemical Physics **87** (1987) p1625-1629

[82] *Relaxation of metastable N_2($a^1\Pi_g$, v'=0-2) in nitrogen afterglow* L. Magne, G. Cernogora and P. Veis J. Phys. D: Appl. Phys. **25** (1992) p.472-476

[83] *Vacuum-Ultraviolet Photolysis of N_2O. IV. Deactivation of N(^2D)* G. Black, T. G. Slanger, G. A. St. John, and R. A. Young J. Chem. Phys **51** (1969) p.116-121

[84] *Rate coefficients for atomic nitrogen(^2D) reactions* Lawrence G. Piper, Megan E. Donahue, Wilson T. Rawlins J. Phys. Chem. **91** (1987) p3883–3888

[85] *Recent advances in the chemistry of electronically excited atoms* R.J. Donovan, D. Husain Chem. Rev. **70** (1970) pp 489–516

[86] *The excitation and quenching of N(^2P)* R. A. Young, and O. J. Dunn The Journal of Chemical Physics **63** (1975) p.1150-1153

[87] *A $^3\Sigma^+_u$ molecules in the N_2 afterglow* D.E. Shemansky J. Chem. Phys. **64** (1976) p565-580

[88] Delcroix J.L., Ferreira C.M. and Ricard A. (1976), in Principles of Laser Plasmas, ed. G. Bekefi, John Wiley and Sons, New York

[89] Krivonosova O.E., Losev S.A., Nalivaiko V.P., Mukosev Yu.K. and Shatalov O.P. (1987), in Plasma Chemistry, ed. B.M. Smirnov, Energoatomizdat, Moscow, Vol. 14, p. 3 (in Russian)

[90] *NIST Chemical Kinetics Database Standard Reference Database 17, Version 7.0 (Web Version), Release 1.6.8 Data Version 2016.10* kinetics.nist.gov/

[91] *Evaluated Kinetic Data for Combustion Modeling: Supplement II* D. L. Baulch S, C. T. Bowman , C. J. Cobos, R. A. Cox , Th. Just,,J. A. Kerr , M. J. Pilling , D. Stocker , J. Troe , W. Tsang , R. W. Walker , J. Warnatz J. Phys. Chem. Ref. Data **34** (2005) p. 757-1397

[92] *Reactions of N(^{4}S) atoms with NO and H_2* M. Koshi, M. Yoshimura, K. Fukuda, H. Matsui J. Chem. Phys. **93** (1990) p.8703 - 8708

[93] *Mechanism and modeling of nitrogen chemistry in combustion* Miller J A and Bowman C T Prog. Energy and Combust. Sci. **15** (1989) p.287-338

[94] *A pyrolysis mechanism for ammonia* D. F. Davidson, K. Kohse-Höinghaus, A. Y. Chang and R. K. Hanson Int. J.Chem. Kinet. **22** (1990) p.513-535

[95] *The kinetics and thermodynamics of the reaction H+NH3⇄NH2+H2 by the flash photolysis–shock tube technique: Determination of the equilibrium constant, the rate constant for the back reaction, and the enthalpy of formation of the amidogen radical* J. W. Sutherland, and J. V. Michael J. Chem. Phys. **88** (1988) p.830

[96] *Reaction kinetics of NH in the shock tube pyrolysis of HNCO* Mertens J D, Chang A Y, Hanson R K and Bowman C T Int. J.Chem. Kinet. **21** (1989) p.1049-1067

[97] *The O + NH_3 reaction: A review* N. Cohen Int. J.Chem. Kinet. **19** (1987) p 319–362

[98] *A Shock-tube Study of Ammonia Pyrolysis* J.E. Dove, W.S. Nip Can. J. Chem. **57** (1979) p.689-701

[99] M. A. Lieberman and A. J. Lichtenberg, Principles of plasma discharges and materials processing (John Wiley and Sons, Inc., Hoboken, New Jersey, 2005.

[100] *Elementary Steps in Ammonia Synthesis : The surface science approach* G. Ertl in Fundamental and Applied Catalysis : Catalytic Ammonia Synthesis (1991) p 109-132

[101] *Iron single crystals as ammonia synthesis catalysts: Effect of surface structure on catalyst activity* N.D. Spencer, R.C. Schoonmaker, G.A. Somorjai Journal of Catalysis, **74**(1) (1982) Pages 129-135

[102] *Plasma Chemical Synthesis. I. Effect of Electrode Material on the Synthesis of Ammonia* Khin Swe Yin I and Mundiyath Venugopalan Plasma Chemistry and Plasma Processing, Vol**. 3** (1983) p343-350

[103] *Reactions at Surfaces: From Atoms to Complexity (Nobel Lecture* Gerhard Ertl Angew. Chem. Int. **47** (2008) p3524 – 3535

[104] *Transport of adsorbates at metal surfaces: from thermal migration to hot precursors* J.V. Barth Surface Science Reports **40** (2000) p.75-149

[105] *Proton transfer chains in cold plasmas of H_2 with small amounts of N_2. The prevalence of NH_4^+ Esther Carrasco, Isabel Tanarro, Vıctor J. Herrero and José Cernicharo* Phys. Chem. Chem. Phys. **15** (2013) p.1699-1706

[106] *Isotopic exchange processes in cold plasmas of H_2/D_2 mixtures* M. Jiménez-Redondo, E. Carrasco, V. J. Herrero and I. Tanarro Phys. Chem. Chem. Phys. **13** (2011) p.9655-9666

[107] *Surface Science and Catalysis—Studies on the Mechanism of Ammonia Synthesis: The P. H. Emmett Award Address* G. Ertl, Catal. Rev. Sci. Eng. **21** (1980) p.201-223

[108] *Kinetic Simulation of Ammonia Synthesis Catalysis* J. A. Dumesic and *A. A.* Treviao J. of Catalysis **116** (1989) p.119-129

[109] *NH_3 and $NH_{x<3}$ radicals synthesis downstream a microwave discharge sustained in an Ar-N_2-H_2 gas mixture. Study of surface reactive processes and determination of rate constants* J. L. Jauberteau, I Jauberteau and J Aubreton Journal of Physics D: Applied Physics **35** (2002) p.665-675